

IIT JAM Mathematics (MA)

Topic-Wise Question Bank (2022 – 2026)

Categorized by Official Syllabus with Complete Master Answer Key

Contents

1 Real Analysis	1
1.1 Sequences and Series of Real Numbers	1
1.2 Functions of One Real Variable	10
2 Multivariable Calculus and Differential Equations	19
2.1 Functions of Two or Three Real Variables	19
2.2 Integral Calculus	23
2.3 Differential Equations	27
3 Linear Algebra and Algebra	34
3.1 Basic Algebra	34
3.2 Matrices and Systems of Linear Equations	34
3.3 Finite Dimensional Vector Spaces	40
3.4 Groups	45
4 Master Answer Key (2022 – 2026)	52

1 Real Analysis

1.1 Sequences and Series of Real Numbers

JAM 2026	Q.1	MCQ	[1 Mark]
----------	-----	-----	----------

For each positive integer n , let $x_n = 1 - (-1)^n + \frac{1}{n}$ and $y_n = \left(1 + \frac{1}{2n}\right)^{3n}$. Which ONE of the following statements about the sequences (x_n) and (y_n) is TRUE?

- (A) (x_n) and (y_n) are convergent.
- (B) (x_n) is not convergent and (y_n) is not convergent.
- (C) (x_n) is convergent and (y_n) is not convergent.
- (D) (x_n) is not convergent and (y_n) is convergent.

JAM 2026	Q.2	MCQ	[1 Mark]
----------	-----	-----	----------

For each positive integer n , define $x_n = (-1)^n$. Which ONE of the following statements about the sequence (x_n) is FALSE?

- (A) There exists $\epsilon > 0$ such that $|x_n - 1| < \epsilon$ for all positive integers n .
- (B) There exists $\epsilon > 0$ such that for all $M > 0$ there exists a positive integer $N > M$ for which $|x_N - 1| > \epsilon$.
- (C) For all $\epsilon > 0$ and $M > 0$ there exists a positive integer N such that $N > M$ and $|x_N - 1| < \epsilon$.
- (D) For all $\epsilon > 0$ and $M > 0$ there exists a positive integer N such that $N > M$ and $|x_N - 1| > \epsilon$.

JAM 2026**Q.11****MCQ****[2 Marks]**

Given a sequence of real numbers (a_n) , define $b_n = \begin{cases} a_n, & \text{if } a_n \geq 0 \\ 0, & \text{if } a_n < 0 \end{cases}$ and $c_n = \begin{cases} a_n, & \text{if } a_n < 0 \\ 0, & \text{if } a_n \geq 0 \end{cases}$ for each positive integer n . Which ONE of the following statements is TRUE?

- (A) If $\sum_{n=1}^{\infty} a_n$ does not converge, then $\sum_{n=1}^{\infty} b_n$ does not converge.
 (B) If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} |a_n|$ does not converge, then $\sum_{n=1}^{\infty} b_n$ converges.
 (C) If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} |a_n|$ does not converge, then $\sum_{n=1}^{\infty} b_n$ does not converge.
 (D) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ converges or $\sum_{n=1}^{\infty} c_n$ converges.

JAM 2026**Q.12****MCQ****[2 Marks]**

For which one of the following pairs of values of a and b , does the series $\sum_{k=1}^{\infty} \frac{(3x)^k}{2\sqrt{k}(1-x)^k}$ converge for all $x \in (a, b)$?

- (A) $a = -\frac{1}{4}$ and $b = \frac{1}{2}$
 (B) $a = -1$ and $b = \frac{1}{5}$
 (C) $a = -\frac{1}{2}$ and $b = \frac{1}{4}$
 (D) $a = -\frac{1}{4}$ and $b = \frac{1}{3}$

JAM 2026**Q.29****MCQ****[2 Marks]**

For each positive integer n , let $x_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log_e n$ and $y_n = \int_1^n \frac{\cos t}{t^2} dt$. Which ONE of the following statements about (x_n) and (y_n) is TRUE?

- (A) (x_n) and (y_n) are convergent.
 (B) (x_n) is convergent and (y_n) is not convergent.
 (C) (x_n) is not convergent and (y_n) is convergent.
 (D) (x_n) is not convergent and (y_n) is not convergent.

JAM 2026**Q.32****MSQ****[2 Marks]**

Which of the following statements is/are TRUE?

- (A) There exists a monotone sequence that does not converge but has a convergent subsequence.
 (B) There exists a sequence that has a bounded subsequence but does not have any convergent subsequence.
 (C) There exists a sequence (x_n) such that given any positive integer m , (x_n) has a subsequence converging to m .
 (D) There exists a sequence (x_n) such that $(|x_{n+1} - x_n|)$ converges to 0 but (x_n) does not converge.

JAM 2026**Q.41****NAT****[1 Mark]**

The radius of convergence of the series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} (\log_e n)^{-1} x^n$ is _____ (rounded off to one decimal place).

JAM 2026**Q.42****NAT****[1 Mark]**

$$\int_0^1 \left(\sum_{k=1}^{\infty} \frac{(\log_e 2)^k x^{k-1}}{(k-1)!} \right) dx = \text{_____} \text{ (rounded off to one decimal place).}$$

JAM 2026

Q.43

NAT

[1 Mark]

$$\lim_{n \rightarrow \infty} \left(\frac{n^2+1}{\sqrt{n^6+1}} + \cdots + \frac{n^2+n}{\sqrt{n^6+n}} \right) = \text{_____} \text{ (rounded off to one decimal place).}$$

JAM 2025

Q.1

MCQ

[1 Mark]

The sum of the infinite series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{\pi^{2n+1}}{2^{2n+1}(2n)!}$ is equal to:

- (A) $-\pi$
- (B) $\frac{\pi}{4}$
- (C) $\frac{\pi}{2}$
- (D) $-\frac{\pi}{4}$

JAM 2025

Q.18

MCQ

[2 Marks]

For $n \in \mathbb{N}$, define $x_n = (-1)^n \cos \frac{1}{n}$ and $y_n = \sum_{k=1}^n \frac{1}{n+k}$. Then, which one of the following is TRUE?

- (A) $\sum_{n=1}^{\infty} x_n$ converges, and $\sum_{n=1}^{\infty} y_n$ does NOT converge
- (B) $\sum_{n=1}^{\infty} x_n$ does NOT converge, and $\sum_{n=1}^{\infty} y_n$ converges
- (C) $\sum_{n=1}^{\infty} x_n$ converges, and $\sum_{n=1}^{\infty} y_n$ converges
- (D) $\sum_{n=1}^{\infty} x_n$ does NOT converge, and $\sum_{n=1}^{\infty} y_n$ does NOT converge

JAM 2025

Q.19

MCQ

[2 Marks]

Let $x_1 = \frac{5}{2}$. For $n \in \mathbb{N}$, define $x_{n+1} = \frac{1}{5}(x_n^2 + 6)$. Then, which one of the following is TRUE?

- (A) (x_n) is an increasing sequence, and (x_n) is NOT a bounded sequence
- (B) (x_n) is NOT an increasing sequence, and (x_n) is NOT a bounded sequence
- (C) (x_n) is NOT a decreasing sequence, and (x_n) is a bounded sequence
- (D) (x_n) is a decreasing sequence, and (x_n) is a bounded sequence

JAM 2025

Q.20

MCQ

[2 Marks]

Let $x_1 = 2$ and $x_{n+1} = 2 + \frac{1}{2x_n}$ for all $n \in \mathbb{N}$. Then, which one of the following is TRUE?

- (A) $x_{n+1} \geq \frac{4}{x_n}$ for all $n \in \mathbb{N}$, and (x_n) is a Cauchy sequence
- (B) $x_{n+1} < \frac{4}{x_n}$ for some $n \in \mathbb{N}$, and (x_n) is a Cauchy sequence
- (C) $x_{n+1} \geq \frac{4}{x_n}$ for all $n \in \mathbb{N}$, and (x_n) is NOT a Cauchy sequence
- (D) $x_{n+1} < \frac{4}{x_n}$ for some $n \in \mathbb{N}$, and (x_n) is NOT a Cauchy sequence

JAM 2025

Q.21

MCQ

[2 Marks]

For $n \in \mathbb{N}$, define $x_n = (-1)^n \frac{3^n}{n^3}$ and $y_n = (4^n + (-1)^n 3^n)^{1/n}$. Then, which one of the following is TRUE?

- (A) (x_n) has a convergent subsequence, and NO subsequence of (y_n) is convergent

- (B) NO subsequence of (x_n) is convergent, and (y_n) has a convergent subsequence
- (C) (x_n) has a convergent subsequence, and (y_n) has a convergent subsequence
- (D) NO subsequence of (x_n) is convergent, and NO subsequence of (y_n) is convergent

JAM 2025

Q.25

MCQ

[2 Marks]

Let $x_1 = 1$. For $n \in \mathbb{N}$, define $x_{n+1} = \left(\frac{1}{2} + \frac{\sin^2 n}{n}\right) x_n$. Then, which one of the following is TRUE?

- (A) $\sum_{n=1}^{\infty} x_n$ converges
- (B) $\sum_{n=1}^{\infty} x_n$ does NOT converge
- (C) $\sum_{n=1}^{\infty} x_n^2$ does NOT converge
- (D) $\sum_{n=1}^{\infty} x_n x_{n+1}$ does NOT converge

JAM 2025

Q.26

MCQ

[2 Marks]

Let $x_1 > 0$. For $n \in \mathbb{N}$, define $x_{n+1} = x_n + 4$. If $\lim_{n \rightarrow \infty} \left(\frac{1}{x_2 x_3} + \frac{1}{x_3 x_4} + \cdots + \frac{1}{x_{n+1} x_{n+2}} \right) = \frac{1}{24}$, then the value of x_1 is equal to:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

JAM 2025

Q.36

MSQ

[2 Marks]

For $n \in \mathbb{N}$, define $x_n = \sum_{k=1}^n \frac{k}{n^2+k}$. Then, which of the following is/are TRUE?

- (A) The sequence (x_n) converges
- (B) The series $\sum_{n=1}^{\infty} x_n$ converges
- (C) The series $\sum_{n=1}^{\infty} x_n$ does NOT converge
- (D) The series $\sum_{n=1}^{\infty} \frac{x_n}{n}$ converges

JAM 2025

Q.41

NAT

[1 Mark]

The radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{(x+\frac{1}{4})^n}{(-2)^n n^2}$ about $x = -\frac{1}{4}$ is equal to _____ (rounded off to two decimal places).

JAM 2025

Q.54

NAT

[2 Marks]

$\sum_{n=1}^{\infty} n \left(\frac{3}{4}\right)^{2(n-1)} = \text{_____}$ (rounded off to two decimal places).

JAM 2024

Q.14

MCQ

[2 Marks]

For $n \in \mathbb{N}$, let $a_n = \frac{1}{(3n+2)(3n+4)}$ and $b_n = \frac{n^3 + \cos(3^n)}{3^n + n^3}$. Then, which one of the following is TRUE?

- (A) $\sum_{n=1}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} b_n$ is divergent
- (B) $\sum_{n=1}^{\infty} a_n$ is divergent but $\sum_{n=1}^{\infty} b_n$ is convergent
- (C) Both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are divergent

(D) Both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent

JAM 2024

Q.20

MCQ

[2 Marks]

Define the sequences $\{a_n\}_{n=3}^{\infty}$ and $\{b_n\}_{n=3}^{\infty}$ as $a_n = (\log n + \log \log n)^{\log n}$ and $b_n = n^{(1+\frac{1}{\log n})}$. Which one of the following is TRUE?

- (A) $\sum_{n=3}^{\infty} \frac{1}{a_n}$ is convergent but $\sum_{n=3}^{\infty} \frac{1}{b_n}$ is divergent
- (B) $\sum_{n=3}^{\infty} \frac{1}{a_n}$ is divergent but $\sum_{n=3}^{\infty} \frac{1}{b_n}$ is convergent
- (C) Both $\sum_{n=3}^{\infty} \frac{1}{a_n}$ and $\sum_{n=3}^{\infty} \frac{1}{b_n}$ are divergent
- (D) Both $\sum_{n=3}^{\infty} \frac{1}{a_n}$ and $\sum_{n=3}^{\infty} \frac{1}{b_n}$ are convergent

JAM 2024

Q.21

MCQ

[2 Marks]

For $p, q, r \in \mathbb{R}$, $r \neq 0$ and $n \in \mathbb{N}$, let $a_n = p^n n^q \left(\frac{n}{n+2}\right)^{n^2}$ and $b_n = \frac{n^n}{n! r^n} \left(\sqrt{\frac{n+2}{n}}\right)$. Then:

- (A) If $1 < p < e^2$ and $q > 1$, then $\sum_{n=1}^{\infty} a_n$ is convergent
- (B) If $e^2 < p < e^4$ and $q > 1$, then $\sum_{n=1}^{\infty} a_n$ is convergent
- (C) If $1 < r < e$, then $\sum_{n=1}^{\infty} b_n$ is convergent
- (D) If $\frac{1}{e} < r < 1$, then $\sum_{n=1}^{\infty} b_n$ is convergent

JAM 2024

Q.28

MCQ

[2 Marks]

For $n \geq 3$, let a regular n -sided polygon P_n be circumscribed by a circle of radius R_n and let r_n be the radius of the circle inscribed in P_n . Then $\lim_{n \rightarrow \infty} \left(\frac{R_n}{r_n}\right)^{n^2}$ equals:

- (A) $e^{(\pi^2)}$
- (B) $e^{(\frac{\pi^2}{2})}$
- (C) $e^{(\frac{\pi^2}{3})}$
- (D) $e^{(2\pi^2)}$

JAM 2024

Q.31

MSQ

[2 Marks]

Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers. Then, which of the following statements is/are always TRUE?

- (A) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n^2$ converges absolutely
- (B) If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n^3$ converges absolutely
- (C) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^2$ converges
- (D) If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} a_n^3$ converges

JAM 2024

Q.32

MSQ

[2 Marks]

Which of the following statements is/are TRUE?

- (A) $\sum_{n=1}^{\infty} n \log(1 + \frac{1}{n^3})$ is convergent

- (B) $\sum_{n=1}^{\infty} (1 - \cos(\frac{1}{n})) \log n$ is convergent
- (C) $\sum_{n=1}^{\infty} n^2 \log(1 + \frac{1}{n^3})$ is convergent
- (D) $\sum_{n=1}^{\infty} (1 - \cos(\frac{1}{\sqrt{n}})) \log n$ is convergent

JAM 2024

Q.39

MSQ

[2 Marks]

For $0 < \alpha < 4$, define the sequence $\{x_n\}_{n=1}^{\infty}$ as $x_1 = \alpha$ and $x_{n+1} + 2 = -x_n(x_n - 4)$ for $n \in \mathbb{N}$. Which of the following is/are TRUE?

- (A) $\{x_n\}$ converges for at least three distinct values of $\alpha \in (0, 1)$
- (B) $\{x_n\}$ converges for at least three distinct values of $\alpha \in (1, 2)$
- (C) $\{x_n\}$ converges for at least three distinct values of $\alpha \in (2, 3)$
- (D) $\{x_n\}$ converges for at least three distinct values of $\alpha \in (3, 4)$

JAM 2024

Q.47

NAT

[1 Mark]

Let $a_1 = 1, b_1 = 2, c_1 = 3$. Define $a_{n+1} = \frac{a_n + b_n}{2}, b_{n+1} = \frac{b_n + c_n}{2}, c_{n+1} = \frac{c_n + a_n}{2}$. Then,

$$\sum_{n=1}^{\infty} b_n c_n (a_{n+1} - a_n) + \sum_{n=1}^{\infty} (b_{n+1} c_{n+1} - b_n c_n) a_{n+1} = \underline{\hspace{2cm}}$$

(rounded off to two decimal places).

JAM 2024

Q.55

NAT

[2 Marks]

If $a_n = \frac{1}{n^3+1} + \frac{2^2}{n^3+2} + \dots + \frac{n^2}{n^3+n}$, then the sequence $\{a_n\}$ converges to _____ (rounded off to two decimal places).

JAM 2024

Q.59

NAT

[2 Marks]

Let $a_n = \frac{1}{n^{n-1}} \sum_{k=0}^n \frac{n!}{k!(n-k)!} \frac{n^k}{k+1}$ and $\beta = \lim_{n \rightarrow \infty} a_n$. Then, the value of $\log \beta$ equals _____ (rounded off to two decimal places).

JAM 2023

Q.9

MCQ

[1 Mark]

Let $a_n = (1 + \frac{1}{n})^n$ and $b_n = n \cos(\frac{n! \pi}{2^{10}})$ for $n \in \mathbb{N}$. Then:

- (A) (a_n) is convergent and (b_n) is bounded
- (B) (a_n) is NOT convergent and (b_n) is bounded
- (C) (a_n) is convergent and (b_n) is unbounded
- (D) (a_n) is NOT convergent and (b_n) is unbounded

JAM 2023

Q.10

MCQ

[1 Mark]

Let (a_n) be a sequence of real numbers defined by $a_n = \begin{cases} 1, & \text{if } n \text{ is prime} \\ -1, & \text{if } n \text{ is not prime} \end{cases}$. Let $b_n = \frac{a_n}{n}$ for $n \in \mathbb{N}$. Then:

- (A) both (a_n) and (b_n) are convergent
- (B) (a_n) is convergent but (b_n) is NOT convergent
- (C) (a_n) is NOT convergent but (b_n) is convergent

(D) both (a_n) and (b_n) are NOT convergent

JAM 2023

Q.11

MCQ

[2 Marks]

Let $a_n = \sin(\frac{1}{n^3})$ and $b_n = \sin(\frac{1}{n})$ for $n \in \mathbb{N}$. Then:

- (A) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent
- (B) $\sum_{n=1}^{\infty} a_n$ is convergent but $\sum_{n=1}^{\infty} b_n$ is NOT convergent
- (C) $\sum_{n=1}^{\infty} a_n$ is NOT convergent but $\sum_{n=1}^{\infty} b_n$ is convergent
- (D) both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are NOT convergent

JAM 2023

Q.19

MCQ

[2 Marks]

Let $a_n = \frac{1+2^{-2}+\dots+n^{-2}}{n}$ for $n \in \mathbb{N}$. Then:

- (A) both (a_n) and $\sum_{n=1}^{\infty} a_n$ are convergent
- (B) (a_n) is convergent but $\sum_{n=1}^{\infty} a_n$ is NOT convergent
- (C) both (a_n) and $\sum_{n=1}^{\infty} a_n$ are NOT convergent
- (D) (a_n) is NOT convergent but $\sum_{n=1}^{\infty} a_n$ is convergent

JAM 2023

Q.20

MCQ

[2 Marks]

Let (a_n) be a sequence of real numbers such that $\sum_{n=0}^{\infty} a_n(x-2)^n$ converges at $x = -5$. Then this series also converges at:

- (A) $x = 9$
- (B) $x = 12$
- (C) $x = 5$
- (D) $x = -6$

JAM 2023

Q.21

MCQ

[2 Marks]

Let (a_n) and (b_n) be sequences of real numbers such that $|a_n - a_{n+1}| = \frac{1}{2^n}$ and $|b_n - b_{n+1}| = \frac{1}{\sqrt{n}}$ for $n \in \mathbb{N}$. Then:

- (A) both (a_n) and (b_n) are Cauchy sequences
- (B) (a_n) is a Cauchy sequence but (b_n) need NOT be a Cauchy sequence
- (C) (a_n) need NOT be a Cauchy sequence but (b_n) is a Cauchy sequence
- (D) both (a_n) and (b_n) need NOT be Cauchy sequences

JAM 2023

Q.24

MCQ

[2 Marks]

Suppose $f : (-1, 1) \rightarrow \mathbb{R}$ is an infinitely differentiable function such that $\sum_{j=0}^{\infty} a_j \frac{x^j}{j!}$ converges to $f(x)$ for each $x \in (-1, 1)$, where $a_j = \int_0^{\pi/2} \theta^j \cos^j(\tan \theta) d\theta + \int_{\pi/2}^{\pi} (\theta - \pi)^j \cos^j(\tan \theta) d\theta$ for $j \geq 0$. Then:

- (A) $f(x) = 0$ for all $x \in (-1, 1)$
- (B) f is a non-constant even function on $(-1, 1)$

- (C) f is a non-constant odd function on $(-1, 1)$
- (D) f is NEITHER an odd function NOR an even function on $(-1, 1)$

JAM 2023

Q.40

MSQ

[2 Marks]

Let R_1 and R_2 be the radii of convergence of $\sum_{n=1}^{\infty} (-1)^n x^{n-1}$ and $\sum_{n=1}^{\infty} (-1)^n \frac{x^{n+1}}{n(n+1)}$, respectively. Then:

- (A) $R_1 = R_2$
- (B) $R_2 > 1$
- (C) $\sum_{n=1}^{\infty} (-1)^n x^{n-1}$ converges for all $x \in [-1, 1]$
- (D) $\sum_{n=1}^{\infty} (-1)^n \frac{x^{n+1}}{n(n+1)}$ converges for all $x \in [-1, 1]$

JAM 2023

Q.49

NAT

[1 Mark]

The sum of the series $\sum_{n=1}^{\infty} \frac{2n+1}{(n^2+1)(n^2+2n+2)}$ is equal to _____ (rounded off to two decimal places).

JAM 2023

Q.50

NAT

[1 Mark]

$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{2^n} + \frac{1}{3^n} + \cdots + \frac{1}{(2023)^n} \right)^{\frac{1}{n}} =$ _____ (rounded off to two decimal places).

JAM 2022

Q.5

MCQ

[1 Mark]

The radius of convergence of the power series $\sum_{n=1}^{\infty} \left(\frac{n^3}{4^n} \right) x^{5n}$ is:

- (A) 4
- (B) $\sqrt[5]{4}$
- (C) $\frac{1}{4}$
- (D) $\frac{1}{\sqrt[5]{4}}$

JAM 2022

Q.6

MCQ

[1 Mark]

Let (x_n) and (y_n) be defined by $x_1 = 1, y_1 = \frac{1}{2}, x_{n+1} = \frac{x_n + y_n}{2}$, and $y_{n+1} = \sqrt{x_n y_n}$ for all $n \in \mathbb{N}$. Then:

- (A) (x_n) is convergent, but (y_n) is not convergent.
- (B) (x_n) is not convergent, but (y_n) is convergent.
- (C) Both (x_n) and (y_n) are convergent and $\lim_{n \rightarrow \infty} x_n > \lim_{n \rightarrow \infty} y_n$.
- (D) Both (x_n) and (y_n) are convergent and $\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n$.

JAM 2022

Q.7

MCQ

[1 Mark]

Suppose $a_n = \frac{3^n+3}{5^n-5}$ and $b_n = \frac{1}{(1+n^2)^{\frac{1}{4}}}$ for $n = 2, 3, 4, \dots$. Then:

- (A) $\sum_{n=2}^{\infty} a_n$ is convergent and $\sum_{n=2}^{\infty} b_n$ is convergent.
- (B) $\sum_{n=2}^{\infty} a_n$ is convergent and $\sum_{n=2}^{\infty} b_n$ is divergent.
- (C) $\sum_{n=2}^{\infty} a_n$ is divergent and $\sum_{n=2}^{\infty} b_n$ is convergent.

(D) $\sum_{n=2}^{\infty} a_n$ is divergent and $\sum_{n=2}^{\infty} b_n$ is divergent.

JAM 2022

Q.8

MCQ

[1 Mark]

Consider the series $\sum_{n=1}^{\infty} \frac{1}{n^m(1+\frac{1}{n^p})}$, where m and p are real numbers. Under which of the following conditions does the above series converge?

- (A) $m > 1$
- (B) $0 < m < 1$ and $p > 1$
- (C) $0 < m \leq 1$ and $0 \leq p \leq 1$
- (D) $m = 1$ and $p > 1$

JAM 2022

Q.12

MCQ

[2 Marks]

Let $T = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$ and $S = 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \dots$. Then:

- (A) $T = S$ and $S \neq 0$
- (B) $2T = S$ and $S \neq 0$
- (C) $T = 2S$ and $S \neq 0$
- (D) $T = S = 0$

JAM 2022

Q.14

MCQ

[2 Marks]

The value of $\lim_{n \rightarrow \infty} \sum_{k=2}^n \frac{\sqrt{n+1} - \sqrt{n}}{k(\ln k)^2}$ is equal to:

- (A) ∞
- (B) 1
- (C) e
- (D) 0

JAM 2022

Q.28

MCQ

[2 Marks]

For $-1 \leq x \leq 1$, if $f(x)$ is the sum of the convergent power series $x + \frac{x^2}{2^2} + \frac{x^3}{3^2} + \dots + \frac{x^n}{n^2} + \dots$, then $f(\frac{1}{2})$ is equal to:

- (A) $\int_0^{1/2} \frac{\ln(1-t)}{t} dt$
- (B) $-\int_0^{1/2} \frac{\ln(1-t)}{t} dt$
- (C) $\int_0^{1/2} t \ln(1+t) dt$
- (D) $\int_0^{1/2} t \ln(1-t) dt$

JAM 2022

Q.38

MSQ

[2 Marks]

Let (x_n) be a sequence of real numbers. Consider $P = \{n \in \mathbb{N} : x_n > x_m \text{ for all } m > n\}$. Then:

- (A) If P is finite, then (x_n) has a monotonically increasing subsequence.
- (B) If P is finite, then no subsequence of (x_n) is monotonically increasing.
- (C) If P is infinite, then (x_n) has a monotonically decreasing subsequence.

(D) If P is infinite, then no subsequence of (x_n) is monotonically decreasing.

JAM 2022

Q.44

NAT

[1 Mark]

The sum of the series $\sum_{n=1}^{\infty} \frac{1}{(4n-3)(4n+1)}$ is equal to _____ (rounded off to two decimal places).

JAM 2022

Q.55

NAT

[2 Marks]

Let r be the radius of convergence of $\frac{1}{3} + \frac{x}{5} + \frac{x^2}{3^2} + \frac{x^3}{5^2} + \frac{x^4}{3^3} + \dots$. Then the value of r^2 is equal to _____ (rounded off to two decimal places).

1.2 Functions of One Real Variable

JAM 2026

Q.3

MCQ

[1 Mark]

Let $f : (1, 4) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) = (f(x) - \pi x)^2 + \pi$ for all $x \in (1, 4)$. Which ONE of the following is a possible value of $f(3) - f(2)$?

- (A) $\pi + \frac{1}{6}$
- (B) $\pi - \frac{1}{6}$
- (C) $\frac{\pi}{2} + \frac{1}{3}$
- (D) $\frac{\pi}{2} + \frac{1}{2}$

JAM 2026

Q.6

MCQ

[1 Mark]

Let $f : (1, 2) \rightarrow \mathbb{R}$ be a continuous function satisfying $\lim_{x \rightarrow 1^+} f(x) = 3$ and $\lim_{x \rightarrow 2^-} f(x) = 3$. Then which ONE of the following is necessarily TRUE?

- (A) f is bounded and f has a maximum or a minimum but not both.
- (B) f is bounded and f has a maximum or a minimum or both.
- (C) f is bounded and f has neither a maximum nor a minimum.
- (D) f is unbounded and f has either no maximum or no minimum.

JAM 2026

Q.7

MCQ

[1 Mark]

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $f(n) = 1$ for all $n \in \mathbb{Z}$. Which ONE of the following statements is necessarily TRUE?

- (A) $f'(n) = 0$ for all $n \in \mathbb{Z}$
- (B) $f'(x) = 0$ for infinitely many $x \in \mathbb{R} \setminus \mathbb{Z}$
- (C) $f''(n) = 0$ for all $n \in \mathbb{Z}$
- (D) $f''(x) = 0$ for infinitely many $x \in \mathbb{R} \setminus \mathbb{Z}$

JAM 2026

Q.13

MCQ

[2 Marks]

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function that has derivatives of all orders. Which ONE of the following statements is FALSE?

- (A) If f is a polynomial in x of degree at most 5, then $f^{(6)}(x) = 0$ for all $x \in \mathbb{R}$.
- (B) If $f^{(6)}(x) = 0$ for all $x \in \mathbb{R}$, then f is a polynomial in x of degree at most 5.

- (C) If $f^{(k)}(a) = 0$ for all $1 \leq k \leq 5$ and $f^{(6)}(a) > 0$ for some $a \in \mathbb{R}$, then $f(x)$ has a local minimum at a .
- (D) If $f^{(k)}(a) = 0$ for all $1 \leq k \leq 6$ and $f^{(7)}(a) < 0$ for some $a \in \mathbb{R}$, then $f(x)$ has a local maximum at a .

JAM 2026

Q.19

MCQ

[2 Marks]

The value of $\lim_{x \rightarrow \infty} \int_0^x e^{-t} |\sin t| dt$ is:

- (A) $\frac{e^\pi + 1}{2(e^\pi - 1)}$
- (B) $\frac{e^\pi - 1}{2(e^\pi + 1)}$
- (C) $\frac{2(e^\pi + 1)}{e^\pi - 1}$
- (D) $\frac{2(e^\pi - 1)}{e^\pi + 1}$

JAM 2026

Q.21

MCQ

[2 Marks]

Let $f : [1, 9] \rightarrow \mathbb{R}$ be a non-constant continuous function such that $f(1) = f(9)$. Which ONE of the following statements is necessarily TRUE?

- (A) There exists $c \in [4, 8]$ such that $f(c) = f(c + 1)$.
- (B) There exists $c \in [3, 7]$ such that $f(c) = f(c + 2)$.
- (C) There exists $c \in [2, 6]$ such that $f(c) = f(c + 3)$.
- (D) There exists $c \in [1, 5]$ such that $f(c) = f(c + 4)$.

JAM 2026

Q.31

MSQ

[2 Marks]

Which of the following functions $f : \mathbb{R} \rightarrow \mathbb{R}$ has/have a local minimum at $x = 0$?

- (A) $f(x) = \sin |x|$
- (B) $f(x) = \sin x + \frac{x^3}{6}$
- (C) $f(x) = x^4 + x^2 + 3$
- (D) $f(x) = \min\{x - [x], 1 - x + [x]\}$

JAM 2026

Q.33

MSQ

[2 Marks]

Which of the following functions is/are differentiable at $x = 1$?

- (A) $f(x) = |x - 1|^3$
- (B) $f(x) = |x^2 - 1|$
- (C) $f(x) = \begin{cases} x^2 e^{-x^2}, & |x| \leq 1 \\ e^{-1}, & |x| > 1 \end{cases}$
- (D) $f(x) = [x]$

JAM 2026

Q.34

MSQ

[2 Marks]

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = \begin{cases} \cos x, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$. Which of the following is/are TRUE?

- (A) $f(x)$ is continuous at 0.
- (B) $f(x)$ is continuous at $\frac{\pi}{2}$.
- (C) $f(x)$ is Riemann integrable on $[0, 1]$ and $\int_0^1 f(x)dx = \sin 1$.
- (D) $f(x)$ is Riemann integrable on $[0, 1]$ and $\int_0^1 f(x)dx = 0$.

JAM 2026 **Q.44** **NAT** **[1 Mark]**

$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} + e^{\frac{1-\cos x}{x}} \right) = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

JAM 2026 **Q.48** **NAT** **[1 Mark]**

$\int_0^3 (|x-1| - x[x]) dx = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

JAM 2026 **Q.54** **NAT** **[2 Marks]**

Let $f(x) = \int_0^{e^{\sin x}} \int_0^{\log_e y} e^{-(\frac{1}{\sqrt{2}})} (1-t^2) dt dy$. Then $f'(\pi/4) = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

JAM 2025 **Q.3** **MCQ** **[1 Mark]**

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} x|x| \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ and $g(x) = \begin{cases} x^2 \sin \frac{1}{x} + x \cos \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$.

Then:

- (A) f is differentiable at $x = 0$, and g is NOT differentiable at $x = 0$
- (B) f is NOT differentiable at $x = 0$, and g is differentiable at $x = 0$
- (C) f is differentiable at $x = 0$ and g is differentiable at $x = 0$
- (D) f is NOT differentiable at $x = 0$, and g is NOT differentiable at $x = 0$

JAM 2025 **Q.4** **MCQ** **[1 Mark]**

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} |x|^{1/3} |\sin \frac{1}{x}| \cos x, & x \neq 0 \\ 0, & x = 0 \end{cases}$ and $g(x) = \begin{cases} e^x \cos \frac{1}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$.

Then:

- (A) f is continuous at $x = 0$, and g is NOT continuous at $x = 0$
- (B) f is NOT continuous at $x = 0$, and g is continuous at $x = 0$
- (C) f is continuous at $x = 0$, and g is continuous at $x = 0$
- (D) f is NOT continuous at $x = 0$, and g is NOT continuous at $x = 0$

JAM 2025 **Q.17** **MCQ** **[2 Marks]**

Let $f(x) = 10x^2 + e^x - \sin(2x) - \cos x, x \in \mathbb{R}$. The number of points at which the function f has a local minimum is:

- (A) 0
- (B) 1
- (C) 2
- (D) greater than or equal to 3

JAM 2025**Q.30****MCQ****[2 Marks]**

The number of elements in the set $\{x \in \mathbb{R} : 8x^2 + x^4 + x^8 = \cos x\}$ is equal to:

- (A) 0
- (B) 1
- (C) 2
- (D) greater than or equal to 3

JAM 2025**Q.37****MSQ****[2 Marks]**

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $f(0) = 0$, $f'(0) = 2$ and $f(1) = -3$. Then, which of the following is/are TRUE?

- (A) $|f'(x)| \leq 2$ for all $x \in [0, 1]$
- (B) $|f'(x_1)| > 2$ for some $x_1 \in [0, 1]$
- (C) $|f''(x)| < 10$ for all $x \in [0, 1]$
- (D) $|f''(x_2)| \geq 10$ for some $x_2 \in [0, 1]$

JAM 2025**Q.38****MSQ****[2 Marks]**

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $f(0) = 4$, $f(1) = -2$, $f(2) = 8$ and $f(3) = 2$. Then, which of the following is/are TRUE?

- (A) $|f'(x)| < 5$ for all $x \in [0, 1]$
- (B) $|f'(x_1)| \geq 5$ for some $x_1 \in [0, 1]$
- (C) $f'(x_2) = 0$ for some $x_2 \in [0, 3]$
- (D) $f''(x_3) = 0$ for some $x_3 \in [0, 3]$

JAM 2025**Q.42****NAT****[1 Mark]**

$\lim_{n \rightarrow \infty} 8n \left(e^{\left(\frac{1}{2n}\right)} - 1 \right) \left(\sin \frac{1}{2n} + \left| \cos \frac{1}{2n} \right| \right) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2025**Q.43****NAT****[1 Mark]**

Let α be the real number such that $\lim_{x \rightarrow 0} \frac{(1 - \cos x)(2^{2+x} - 4)}{x^3} = \alpha \ln 2$. Then, $\alpha = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2025**Q.51****NAT****[2 Marks]**

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous satisfying $\int_0^{\pi/4} (\sin(x)f(x) + \cos(x) \int_0^x f(t)dt) dx = \sqrt{2}$. Then, $\int_0^{\pi/4} f(x) dx = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2025**Q.53****NAT****[2 Marks]**

Let $f(x) = 2x - \sin x$ for $x \in \mathbb{R}$. If $\lim_{x \rightarrow 0} \left(\frac{1}{x} \sum_{i=1}^k i^2 f\left(\frac{x}{i}\right) \right) = 45$, then $k = \underline{\hspace{2cm}}$.

JAM 2024**Q.19****MCQ****[2 Marks]**

Let $f : [-2\pi, 2\pi] \rightarrow \mathbb{R}$ be defined by $f(x) = \sin(\min\{x, x - [x]\})$. Consider the set $S = \{x \in [-2\pi, 2\pi] : f \text{ is discontinuous at } x\}$. Then:

- (A) S has 13 elements
- (B) S has 7 elements
- (C) S is infinite
- (D) S has 6 elements

JAM 2024

Q.24

MCQ

[2 Marks]

Consider the statements:

P: $\exists f, g$ such that f is continuous at $x = 1$, g is discontinuous at $x = 1$, but $g \circ f$ is continuous at $x = 1$.

Q: $\exists f, g$ such that both f, g are discontinuous at $x = 1$, but $g \circ f$ is continuous at $x = 1$.

- (A) Both P and Q are true
- (B) Both false
- (C) P true, Q false
- (D) P false, Q true

JAM 2024

Q.25

MCQ

[2 Marks]

Let $f(x) = \frac{(x^2+1)^2}{x^4+x^2+1}$ for $x \in \mathbb{R}$. Then f has:

- (A) exactly 2 local maxima and 3 local minima
- (B) exactly 3 local maxima and 2 local minima
- (C) exactly 1 local maximum and 2 local minima
- (D) exactly 2 local maxima and 1 local minimum

JAM 2024

Q.29

MCQ

[2 Marks]

Let $L_1 : y = 3x + 2$ and $L_2 : y = 4x + 3$. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is C^4 such that L_1 intersects $y = f(x)$ at 3 distinct points and L_2 intersects $y = f(x)$ at 4 distinct points. Then:

- (A) $\frac{df}{dx}$ does not attain 3 on $\mathbb{R}$
- (B) $\frac{d^2f}{dx^2}$ vanishes at most once
- (C) $\frac{d^3f}{dx^3}$ vanishes at least once
- (D) $\frac{df}{dx}$ does not attain $\frac{7}{2}$

JAM 2024

Q.34

MSQ

[2 Marks]

Let $f : (1, \infty) \rightarrow (0, \infty)$ be a continuous function such that $f(n) = n!$ for all $n \in \mathbb{N}$. Then:

- (A) $\lim_{x \rightarrow \infty} f(x)$ does not exist
- (B) $\lim_{x \rightarrow \infty} f(x) = \infty$
- (C) The set of solutions to $f(x) = 2024$ is finite
- (D) The set of solutions to $f(x) = 2024$ is infinite

JAM 2024**Q.38****MSQ****[2 Marks]**

Define $f(x) = \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m}(m!)^2}$ and $g(x) = \frac{x}{2} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m}(m+1)!m!}$. Let $0 < x_1 < x_2, 0 < x_3 < x_4$ with $f(x_1) = f(x_2) = 0$, $f(x) \neq 0$ on (x_1, x_2) , and $g(x_3) = g(x_4) = 0$, $g(x) \neq 0$ on (x_3, x_4) . Then:

- (A) f does not vanish in (x_3, x_4)
- (B) f vanishes exactly once in (x_3, x_4)
- (C) g does not vanish in (x_1, x_2)
- (D) g vanishes exactly once in (x_1, x_2)

JAM 2024**Q.46****NAT****[1 Mark]**

Let $S = \{f : \mathbb{R} \rightarrow \mathbb{R} : f \text{ is a polynomial and } f(f(x)) = (f(x))^{2024} \text{ for } x \in \mathbb{R}\}$. The number of elements in S is _____.

JAM 2024**Q.52****NAT****[2 Marks]**

$\lim_{t \rightarrow \infty} \left(\left(\log(t^2 + \frac{1}{t^2}) \right)^{-1} \int_1^{\pi t} \frac{\sin^2 5x}{x} dx \right) = \text{_____}$ (rounded off to two decimal places).

JAM 2024**Q.56****NAT****[2 Marks]**

Let $f(x) = x^3 - 4x^2 + 4x - 6$ and $S(c) = \{x \in \mathbb{R} : f(x) = c\}$. Then $|S(-7)| + |S(-5)| + |S(3)| = \text{_____}$.

JAM 2024**Q.60****NAT****[2 Marks]**

Let $f(x) = \sin^{-1} x$. If a_6 denotes the coefficient of x^6 in the Taylor series of $(f(x))^2$ about $x = 0$, then $9a_6 = \text{_____}$ (rounded off to two decimal places).

JAM 2023**Q.5****MCQ****[1 Mark]**

Let $p(x) = x^{57} + 3x^{10} - 21x^3 + x^2 + 21$ and $q(x) = p(x) + \sum_{j=1}^{57} p^{(j)}(x)$ for all $x \in \mathbb{R}$. Then q admits:

- (A) NEITHER a global maximum NOR a global minimum on $\mathbb{R}$
- (B) a global maximum but NOT a global minimum on $\mathbb{R}$
- (C) a global minimum but NOT a global maximum on $\mathbb{R}$
- (D) a global minimum and a global maximum on $\mathbb{R}$

JAM 2023**Q.6****MCQ****[1 Mark]**

The limit $\lim_{x \rightarrow 0} \frac{\int_0^x \sin(t^2) dt}{\int_0^x (\ln(t+1))^2 dt}$ is:

- (A) 0
- (B) 1
- (C) $\frac{\pi}{e}$
- (D) non-existent

JAM 2023**Q.25****MCQ****[2 Marks]**

Let $f(x) = \cos(x)$ and $g(x) = 1 - \frac{x^2}{2}$ for $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Then:

- (A) $f(x) \geq g(x)$ for all $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$

- (B) $f(x) \leq g(x)$ for all $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$
- (C) $f(x) - g(x)$ changes sign exactly once on $(-\frac{\pi}{2}, \frac{\pi}{2})$
- (D) $f(x) - g(x)$ changes sign more than once on $(-\frac{\pi}{2}, \frac{\pi}{2})$

JAM 2023

Q.28

MCQ

[2 Marks]

Let $y : \mathbb{R} \rightarrow \mathbb{R}$ be C^2 such that $y(0) = y(1) = 0$ and $y''(x) + x^2 < 0$ for all $x \in [0, 1]$. Then:

- (A) $y(x) > 0$ for all $x \in (0, 1)$
- (B) $y(x) < 0$ for all $x \in (0, 1)$
- (C) $y(x) = 0$ has exactly one solution in $(0, 1)$
- (D) $y(x) = 0$ has more than one solution in $(0, 1)$

JAM 2023

Q.30

MCQ

[2 Marks]

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an infinitely differentiable function such that f'' has exactly two distinct zeroes. Then:

- (A) f' has at most 3 distinct zeroes
- (B) f' has at least 1 zero
- (C) f has at most 3 distinct zeroes
- (D) f has at least 2 distinct zeroes

JAM 2023

Q.34

MSQ

[2 Marks]

Let $f : (-1, 1) \rightarrow \mathbb{R}$ be differentiable satisfying $f(0) = 0$ and $|f'(x)| \leq M|x|$ for all $x \in (-1, 1)$. Then:

- (A) f' is continuous at $x = 0$
- (B) f' is differentiable at $x = 0$
- (C) ff' is differentiable at $x = 0$
- (D) $(f')^2$ is differentiable at $x = 0$

JAM 2023

Q.35

MSQ

[2 Marks]

Which of the following functions is/are Riemann integrable on $[0, 1]$?

- (A) $f(x) = \int_0^x |\frac{1}{2} - t| dt$
- (B) $f(x) = \begin{cases} x \sin(1/x), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$
- (C) $f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \cap [0, 1] \\ -1, & \text{otherwise} \end{cases}$
- (D) $f(x) = \begin{cases} x, & \text{if } x \in [0, 1) \\ 0, & \text{if } x = 1 \end{cases}$

JAM 2023 **Q.44** **NAT** **[1 Mark]**

$\lim_{n \rightarrow \infty} \left(n \int_0^1 \frac{x^n}{x+1} dx \right) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2023 **Q.47** **NAT** **[1 Mark]**

Let $f(x) = \sqrt[3]{x}$ for $x > 0$, and $\theta(h)$ satisfy $f(3+h) - f(3) = hf'(3 + \theta(h)h)$ for $h \in (-1, 1)$. Then $\lim_{h \rightarrow 0} \theta(h) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2023 **Q.52** **NAT** **[2 Marks]**

The global minimum value of $f(x) = |x-1| + |x-2|^2$ on $\mathbb{R}$ is equal to $\underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2023 **Q.60** **NAT** **[2 Marks]**

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be bijective with $f(x) = \sum_{n=1}^{\infty} a_n x^n$ and $f^{-1}(x) = \sum_{n=1}^{\infty} b_n x^n$. If $a_1 = 2$ and $a_2 = 4$, then $b_1 = \underline{\hspace{2cm}}$.

JAM 2022 **Q.22** **MCQ** **[2 Marks]**

Let $H(x) = \frac{1}{2}(e^x + e^{-x})$ and $f(x) = \int_0^\pi H(x \sin \theta) d\theta$. Then:

- (A) $xf''(x) + f'(x) + xf(x) = 0$
- (B) $xf''(x) - f'(x) + xf(x) = 0$
- (C) $xf''(x) + f'(x) - xf(x) = 0$
- (D) $xf''(x) - f'(x) - xf(x) = 0$

JAM 2022 **Q.27** **MCQ** **[2 Marks]**

Given $\lim_{N \rightarrow \infty} \int_0^N e^{-t^2} dt = \frac{\sqrt{\pi}}{2}$, the value of $\lim_{N \rightarrow \infty} \int_0^N \frac{1}{t^2} (e^{-at^2} - e^{-bt^2}) dt$ ($0 < a < b$) is:

- (A) $\sqrt{\pi}(\sqrt{a} - \sqrt{b})$
- (B) $\sqrt{\pi}(\sqrt{a} + \sqrt{b})$
- (C) $-\sqrt{\pi}(\sqrt{a} + \sqrt{b})$
- (D) $\sqrt{\pi}(\sqrt{b} - \sqrt{a})$

JAM 2022 **Q.29** **MCQ** **[2 Marks]**

For $n \in \mathbb{N}$ and $x \geq 1$, let $f_n(x) = \int_0^\pi (x^2 + (\cos \theta)\sqrt{x^2 - 1})^n d\theta$. Then:

- (A) $f_n(x)$ is not a polynomial in x if n is odd and $n \geq 3$.
- (B) $f_n(x)$ is not a polynomial in x if n is even and $n \geq 4$.
- (C) $f_n(x)$ is a polynomial in x for all $n \in \mathbb{N}$.
- (D) $f_n(x)$ is not a polynomial in x for any $n \geq 3$.

JAM 2022 **Q.32** **MSQ** **[2 Marks]**

Let S be the set of continuous $f : [-1, 1] \rightarrow \mathbb{R}$ that are analytic on $(-1, 1)$ with Taylor series converging to $f(x)$, and $f(1/n) = 0$ for all $n \in \mathbb{N}$. Then:

- (A) $f(0) = 0$ for every $f \in S$.

- (B) $f'(1/2) = 0$ for every $f \in S$.
- (C) There exists $f \in S$ such that $f'(1/2) \neq 0$.
- (D) There exists $f \in S$ such that $f(x) \neq 0$ for some $x \in [-1, 1]$.

JAM 2022

Q.33

MSQ

[2 Marks]

Define $f : [0, 1] \rightarrow [0, 1]$ by $f(x) = \begin{cases} 1/2, & x = 0 \\ 1/n, & x = m/n \text{ in lowest terms} \\ 0, & x \text{ irrational} \end{cases}$ and $g(x) = \begin{cases} 0, & x = 0 \\ 1, & x \in (0, 1] \end{cases}$.

Then:

- (A) f is Riemann integrable on $[0, 1]$.
- (B) g is Riemann integrable on $[0, 1]$.
- (C) $f \circ g$ is Riemann integrable on $[0, 1]$.
- (D) $g \circ f$ is Riemann integrable on $[0, 1]$.

JAM 2022

Q.34

MSQ

[2 Marks]

Let $S = \{f : \mathbb{R} \rightarrow \mathbb{R} : |f(x) - f(y)|^2 \leq |x - y|^3 \text{ for all } x, y \in \mathbb{R}\}$. Then:

- (A) Every function in S is differentiable.
- (B) There exists $f \in S$ such that f is differentiable, but not twice differentiable.
- (C) There exists $f \in S$ such that f is twice differentiable, but not thrice differentiable.
- (D) Every function in S is infinitely differentiable.

JAM 2022

Q.41

NAT

[1 Mark]

$\lim_{n \rightarrow \infty} \left(\frac{1^4 + 2^4 + \dots + n^4}{n^5} + \frac{1}{\sqrt{n}} \left(\frac{1}{\sqrt{n+1}} + \dots + \frac{1}{\sqrt{4n}} \right) \right) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2022

Q.54

NAT

[2 Marks]

Let $f(x) = \begin{cases} (x - \pi)e^{\sin x}, & 0 \leq x \leq \pi/2 \\ xe^{\sin x} + \frac{4}{\pi}, & \pi/2 < x \leq \pi \end{cases}$. Then $\int_0^\pi f(x) dx = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2022

Q.59

NAT

[2 Marks]

Let $f, g : (-1, 1) \rightarrow \mathbb{R}$ be C^3 with $f(x) \neq g(x)$ for $x \neq 0$, and $f(0) = g(0) = \ln 2$, $f'(0) = g'(0) = \pi$, $f''(0) = g''(0) = \pi^2$, $f'''(0) = \pi^9$, $g'''(0) = \pi^3$. Then $\lim_{x \rightarrow 0} \frac{e^{f(x)} - e^{g(x)}}{f(x) - g(x)} = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2022

Q.60

NAT

[2 Marks]

If $\int_0^{x^3+x^2} f(t) dt = x^2$ and $\int_0^{g(x)} t^2 dt = 9(x+1)^3$, then $f(2) + g(2) + 16f(12) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

2 Multivariable Calculus and Differential Equations

2.1 Functions of Two or Three Real Variables

JAM 2026

Q.15

MCQ

[2 Marks]

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by $f(x, y) = (x^2 - 1)^2 + (y^2 - 1)^2$. Which ONE of the following is TRUE?

- (A) f has local maxima at exactly two points.
- (B) f has local minima at exactly two points.
- (C) f has local minima at exactly three points.
- (D) f has exactly four saddle points.

JAM 2026

Q.17

MCQ

[2 Marks]

Let $f(x, y) = \begin{cases} \frac{5x^2y - 4y^3}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$. Which ONE of the following statements is FALSE?

- (A) f is continuous at $(0, 0)$
- (B) $\frac{\partial f}{\partial x}$ at $(0, 1) = 0$
- (C) $\frac{\partial f}{\partial y}$ at $(1, 0) = 5$
- (D) $\frac{\partial f}{\partial y}$ is continuous at $(0, 0)$

JAM 2026

Q.36

MSQ

[2 Marks]

Let $f(x, y) = \begin{cases} \frac{x^2y}{1+x^2} \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$. Which of the following statements is/are TRUE?

- (A) $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y)$ exists.
- (B) $\frac{\partial f}{\partial x}$ is continuous at the point $(0, 1)$.
- (C) $\frac{\partial f}{\partial y}$ is continuous at the point $(0, 1)$.
- (D) $\lim_{(x,y) \rightarrow (0,0)} \frac{f(x,y)}{\sqrt{x^2+y^2}}$ does not exist.

JAM 2026

Q.46

NAT

[1 Mark]

Let $z = \cos(4x + 5y)$, where $x = \frac{\pi}{2} + 2\theta$, $y = -(\frac{\pi}{4} + \theta)$. Then $\frac{dz}{d\theta} \Big|_{\theta=\frac{\pi}{4}} = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

JAM 2026

Q.52

NAT

[2 Marks]

Let $f(x, y) = \begin{cases} 4x^2 \tan^{-1}(\frac{y}{2x}) + y^3 \tan^{-1}(\frac{x}{4y^2}), & xy \neq 0 \\ 0, & \text{otherwise} \end{cases}$. Then $\frac{\partial^2 f}{\partial y \partial x}(0, 0) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2025

Q.27

MCQ

[2 Marks]

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = e^y(x^2 + y^2)$. Then:

- (A) f has 2 local minima

- (B) f has 2 local maxima
 (C) f has 1 local minimum
 (D) f has 1 local maximum

JAM 2025

Q.31

MSQ

[2 Marks]

Let $f(x, y) = \begin{cases} \frac{xy^2+y^5}{x^2+y^4}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$. Then:

- (A) The iterated limits $\lim_{x \rightarrow 0} \lim_{y \rightarrow 0} f(x, y)$ and $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y)$ exist
 (B) Exactly one of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exists at $(0, 0)$
 (C) Both partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist at $(0, 0)$
 (D) f is NOT differentiable at $(0, 0)$

JAM 2025

Q.34

MSQ

[2 Marks]

Let $f(x, y) = \begin{cases} \frac{(x^2+\sin x)y^2}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$. Then:

- (A) $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 1$
 (B) $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$
 (C) f is differentiable at $(0, 0)$
 (D) f is NOT differentiable at $(0, 0)$

JAM 2025

Q.50

NAT

[1 Mark]

Let $f(x, y) = \begin{cases} \frac{(x^2-y^2)xy}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$. Then $\frac{\partial f}{\partial y}(1, 0) - \frac{\partial f}{\partial x}(0, 2) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2024

Q.2

MCQ

[1 Mark]

For a C^2 function $g : \mathbb{R} \rightarrow \mathbb{R}$, define $u_g(x, y) = \frac{1}{y} \int_{-y}^y g(x+t)dt$ for $y > 0$. Then:

- (A) $\frac{\partial^2 u_g}{\partial x^2} = \frac{2}{y} \frac{\partial u_g}{\partial y} + \frac{\partial^2 u_g}{\partial y^2}$
 (B) $\frac{\partial^2 u_g}{\partial x^2} = \frac{1}{y} \frac{\partial u_g}{\partial y} + \frac{\partial^2 u_g}{\partial y^2}$
 (C) $\frac{\partial^2 u_g}{\partial x^2} = \frac{2}{y} \frac{\partial u_g}{\partial y} - \frac{\partial^2 u_g}{\partial y^2}$
 (D) $\frac{\partial^2 u_g}{\partial x^2} = \frac{1}{y} \frac{\partial u_g}{\partial y} - \frac{\partial^2 u_g}{\partial y^2}$

JAM 2024

Q.30

MCQ

[2 Marks]

Define $f(x, y) = 12xye^{-(2x+3y-2)}$. If (a, b) is the local maximum point, then $f(a, b)$ equals:

- (A) 2
 (B) 6
 (C) 12

(D) 0

JAM 2024

Q.35

MSQ

[2 Marks]

Let $S = \{(x, y) \in \mathbb{R}^2 : x > 0, y > 0\}$ and $f(x, y) = 2x^2 + 3y^2 - \log x - \frac{1}{6} \log y$. Then:

- (A) Unique local maximum point in S
- (B) Unique local minimum point in S
- (C) For each $(x_0, y_0) \in S$, the set $\{(x, y) \in S : f(x, y) = f(x_0, y_0)\}$ is bounded
- (D) For each $(x_0, y_0) \in S$, the set $\{(x, y) \in S : f(x, y) = f(x_0, y_0)\}$ is unbounded

JAM 2023

Q.8

MCQ

[1 Mark]

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $f(x, y) = (e^x \cos y, e^x \sin y)$. Then the number of points in $\mathbb{R}^2$ that do NOT lie in the range of f is:

- (A) 0
- (B) 1
- (C) 2
- (D) infinite

JAM 2023

Q.16

MCQ

[2 Marks]

Let $f(x, y) = e^{x^2+y^2}$ for $(x, y) \in \mathbb{R}^2$, and a_n be the determinant of the matrix $\begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}$ evaluated at $(\cos n, \sin n)$. Then the limit $\lim_{n \rightarrow \infty} a_n$ is:

- (A) non-existent
- (B) 0
- (C) $6e^2$
- (D) $12e^2$

JAM 2023

Q.17

MCQ

[2 Marks]

Let $f(x, y) = \ln(1 + x^2 + y^2)$. Let $P = f_{xx}, Q = f_{xy}, R = f_{yx}, S = f_{yy}$ at $(0, 0)$. Then:

- (A) $PS - QR > 0$ and $P < 0$
- (B) $PS - QR > 0$ and $P > 0$
- (C) $PS - QR < 0$ and $P > 0$
- (D) $PS - QR < 0$ and $P < 0$

JAM 2023

Q.36

MSQ

[2 Marks]

Which of the following subsets of $\mathbb{R}^2$ is/are bounded?

- (A) $\{(x, y) \in \mathbb{R}^2 : e^{x^2} + y^2 \leq 4\}$
- (B) $\{(x, y) \in \mathbb{R}^2 : |x| + |y| \leq 4\}$
- (C) $\{(x, y) \in \mathbb{R}^2 : x^4 + y^2 \leq 4\}$

(D) $\{(x, y) \in \mathbb{R}^2 : e^{x^3} + y^2 \leq 4\}$

JAM 2023

Q.37

MSQ

[2 Marks]

Let $f(x, y) = \begin{cases} \frac{x^4 y^3}{x^6 + y^6}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0) \end{cases}$. Then:

(A) $\lim_{t \rightarrow 0} \frac{f(t, t) - f(0, 0)}{t}$ exists and equals $\frac{1}{2}$

(B) $\left. \frac{\partial f}{\partial x} \right|_{(0, 0)}$ exists and equals 0

(C) $\left. \frac{\partial f}{\partial y} \right|_{(0, 0)}$ exists and equals 0

(D) $\lim_{t \rightarrow 0} \frac{f(t, 2t) - f(0, 0)}{t}$ exists and equals $\frac{1}{3}$

JAM 2023

Q.41

NAT

[1 Mark]

Let $f(x, y) = \begin{cases} (x^2 - 1)^2 \cos^2\left(\frac{y^2}{x^2 - 1}\right), & x \neq \pm 1 \\ 0, & x = \pm 1 \end{cases}$. The number of points of discontinuity of $f(x, y)$ is _____.

JAM 2023

Q.51

NAT

[2 Marks]

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be defined as $f(x, y, z) = x^3 + y^3 + z^3$, and let $L : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the linear map satisfying

$$\lim_{(x, y, z) \rightarrow (0, 0, 0)} \frac{f(1+x, 1+y, 1+z) - f(1, 1, 1) - L(x, y, z)}{\sqrt{x^2 + y^2 + z^2}} = 0.$$

Then $L(1, 2, 4)$ is equal to _____ (rounded off to two decimal places).

JAM 2022

Q.9

MCQ

[1 Mark]

Let $u(x, t) = \frac{1}{2c} \int_{x-ct}^{x+ct} e^{s^2} ds$ on $\mathbb{R}^2$. Then:

(A) $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

(B) $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

(C) $\frac{\partial u}{\partial t} \frac{\partial u}{\partial x} = 0$

(D) $\frac{\partial^2 u}{\partial t \partial x} = 0$

JAM 2022

Q.10

MCQ

[1 Mark]

Let $u(x, y) = x - \frac{x}{x^2 + y^2}$ and $v(x, y) = y + \frac{y}{x^2 + y^2}$. The value of the Jacobian determinant $\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix}$ at $(\cos \theta, \sin \theta)$ ($\theta \in (\pi/4, \pi/2)$) is:

(A) $4 \sin \theta$

(B) $4 \cos \theta$

(C) $4 \sin^2 \theta$

(D) $4 \cos^2 \theta$

JAM 2022

Q.15

MCQ

[2 Marks]

For $t \in \mathbb{R}$, let $[t]$ denote the floor function. Define $h(x, y) = \begin{cases} \frac{-1}{x^2 - y}, & x^2 \neq y \\ 0, & x^2 = y \end{cases}$ and $g(x) =$

$\begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$. Which ONE is FALSE?

- (A) $\lim_{(x,y) \rightarrow (\sqrt{2}, \pi)} \cos\left(\frac{x^2 y}{x^2 + 1}\right) = -\frac{1}{2}$
- (B) $\lim_{(x,y) \rightarrow (\sqrt{2}, 2)} e^{h(x,y)} = 0$
- (C) $\lim_{(x,y) \rightarrow (e,e)} \ln(x^{y-[y]}) = e - 2$
- (D) $\lim_{(x,y) \rightarrow (0,0)} e^{2y} g(x) = 1$

JAM 2022

Q.36

MSQ

[2 Marks]

Let M be a positive real number and let $u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$ be continuous functions satisfying $\sqrt{(u(x,y))^2 + (v(x,y))^2} \geq M\sqrt{x^2 + y^2}$ for all $(x,y) \in \mathbb{R}^2$. Let $F : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $F(x,y) = (u(x,y), v(x,y))$. Then which of the following is/are true?

- (A) F is injective.
- (B) If K is open in $\mathbb{R}^2$, then $F(K)$ is open in $\mathbb{R}^2$.
- (C) If K is closed in $\mathbb{R}^2$, then $F(K)$ is closed in $\mathbb{R}^2$.
- (D) If E is closed and bounded in $\mathbb{R}^2$, then $F^{-1}(E)$ is closed and bounded in $\mathbb{R}^2$.

JAM 2022

Q.42

NAT

[1 Mark]

Let $u(x_1, x_2, x_3) = x_1 x_2^4 x_3^2 - x_1^3 x_3^4 - 26x_1^2 x_2^2 x_3^3$. Let $x_1 \frac{\partial u}{\partial x_2} + 2x_2 \frac{\partial u}{\partial x_3}$ at (t, t^2, t^3) be ct^k . Then $k =$ _____.

JAM 2022

Q.51

NAT

[2 Marks]

Let $D = \{(x,y) \in \mathbb{R}^2 : x^2 + y^2 < 4\}$. Define $f(x,y) = [x^2 + y^2] \frac{x^2 y^2}{x^4 + y^4}$ and $g(x,y) = [y^2] \frac{xy}{x^2 + y^2}$ with $f(0,0) = g(0,0) = 0$. The number of points in D at which both f and g are discontinuous is _____.

JAM 2022

Q.56

NAT

[2 Marks]

Let $f(x,y) = x^2 + 2y^2 - x$. Let $D = \{(x,y) : x^2 + y^2 \leq 1\}$ and $E = \{(x,y) : \frac{x^2}{4} + \frac{y^2}{9} \leq 1\}$. The total number of elements in $D_{\max} \cup D_{\min} \cup E_{\max} \cup E_{\min}$ is _____.

2.2 Integral Calculus

JAM 2026

Q.5

MCQ

[1 Mark]

Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be continuous. Then $\int_0^2 \int_0^x \int_0^y f(x,y,z) dz dy dx =$:

- (A) $\int_0^2 \int_0^y \int_y^2 f(x,y,z) dx dz dy$
- (B) $\int_0^2 \int_0^x \int_z^2 f(x,y,z) dy dz dx$
- (C) $\int_0^2 \int_y^2 \int_0^x f(x,y,z) dz dx dy$
- (D) $\int_0^2 \int_z^2 \int_0^x f(x,y,z) dy dx dz$

JAM 2026

Q.10

MCQ

[1 Mark]

Let T be the triangular region in the plane with vertices $(0,0), (1,0), (1,1)$. Double integral of f over T in polar coordinates equals:

- (A) $\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f d\theta \right) r dr + \int_1^{\sqrt{2}} \left(\int_0^{\cos^{-1}(1/r)} f d\theta \right) r dr$
- (B) $\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f d\theta \right) r dr - \int_1^{\sqrt{2}} \left(\int_0^{\cos^{-1}(1/r)} f d\theta \right) r dr$

- (C) $\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr + \int_1^{\sqrt{2}} \left(\int_0^{\sin^{-1}(1/r)} f \, d\theta \right) r \, dr$
- (D) $\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr - \int_1^{\sqrt{2}} \left(\int_0^{\sin^{-1}(1/r)} f \, d\theta \right) r \, dr$

JAM 2026

Q.20

MCQ

[2 Marks]

The semi-circle $y = \sqrt{6x - 5 - x^2}$ is rotated clockwise by $\frac{\pi}{2}$ in the plane about the origin. The area of the planar region traced out is:

- (A) 6π
- (B) 4π
- (C) 2π
- (D) π

JAM 2026

Q.22

MCQ

[2 Marks]

Let S be the solid intersection of $S_1 : x^2 + y^2 + (z - 1)^2 \leq 4$ and $S_2 : x^2 + y^2 + (z + 1)^2 \leq 4$. The volume of S is:

- (A) $8 \int_0^{\sqrt{3}} \int_0^{\sqrt{3-x^2}} (\sqrt{4-x^2-y^2} - 1) \, dy \, dx$
- (B) $8 \int_0^2 \int_0^{\sqrt{4-x^2}} (\sqrt{4-x^2-y^2} - 1) \, dy \, dx$
- (C) $2 \int_0^2 \int_0^{\sqrt{4-x^2}} (\sqrt{4-x^2-y^2} - 1) \, dy \, dx$
- (D) $2 \int_0^{\sqrt{3}} \int_0^{\sqrt{3-x^2}} (\sqrt{4-x^2-y^2} - 1) \, dy \, dx$

JAM 2026

Q.53

NAT

[2 Marks]

The double integral of $f(x, y) = x$ over the triangular region with vertices $(-\frac{1}{2}, \frac{1}{2})$, $(1, 2)$ and $(1, -1)$ is _____ (rounded off to one decimal place).

JAM 2026

Q.55

NAT

[2 Marks]

The volume of the tetrahedron bounded by $x = 1, y = 2, z = 3$ and $12x + 8y + 6z = 70$ is _____ (rounded off to one decimal place).

JAM 2025

Q.10

MCQ

[1 Mark]

The value of $\int_0^1 \left(\int_{\sqrt{y}}^1 3e^{x^3} \, dx \right) dy$ is equal to:

- (A) $e - 1$
- (B) $\frac{e-1}{2}$
- (C) $\sqrt{e} - 1$
- (D) $\frac{\sqrt{e}-1}{2}$

JAM 2025

Q.28

MCQ

[2 Marks]

Let Ω be the bounded region in $\mathbb{R}^3$ in the first octant bounded by $z = x^2 + y^2, z = 4, x = 0, y = 0$. The volume of Ω is:

- (A) π

- (B) 2π
 (C) 3π
 (D) 4π

JAM 2025 **Q.45** **NAT** **[1 Mark]**

Let S be the surface area of the portion of the plane $z = x + y + 3$ inside $x^2 + y^2 = 1$. Then $(\frac{S}{\pi})^2 =$ _____ (rounded off to two decimal places).

JAM 2025 **Q.48** **NAT** **[1 Mark]**

Let T denote the triangle bounded by the x -axis, $y = x$ and $x = 1$. Then $\iint_T (5 - y) dx dy =$ _____ (rounded off to two decimal places).

JAM 2025 **Q.57** **NAT** **[2 Marks]**

Let Ω be the solid bounded by $z = 0, y = 0, x = \frac{1}{2}, 2y = x$ and $2x + y + z = 4$. If V is the volume, then $64V =$ _____.

JAM 2024 **Q.27** **MCQ** **[2 Marks]**

For $a > b > 0$, let $D = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq a^2 \text{ and } x^2 + y^2 \geq b^2\}$. The surface area of the boundary of D is:

- (A) $4\pi(a + b)\sqrt{a^2 - b^2}$
 (B) $4\pi(a^2 - b\sqrt{a^2 - b^2})$
 (C) $4\pi(a - b)\sqrt{a^2 - b^2}$
 (D) $4\pi(a^2 + b\sqrt{a^2 - b^2})$

JAM 2024 **Q.41** **NAT** **[1 Mark]**

The area of the region $R = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 \leq y \leq 1, \frac{1}{4} \leq xy \leq \frac{1}{2}\}$ is _____ (rounded off to two decimal places).

JAM 2024 **Q.48** **NAT** **[1 Mark]**

Let $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 4, (x - 1)^2 + y^2 \leq 1, z \geq 0\}$. The surface area of S equals _____ (rounded off to two decimal places).

JAM 2024 **Q.53** **NAT** **[2 Marks]**

Let T be the square region enclosed by $(0, 1), (1, 0), (0, -1), (-1, 0)$. Then $\iint_T (\cos(\pi(x - y)) - \cos(\pi(x + y)))^2 dx dy =$ _____ (rounded off to two decimal places).

JAM 2024 **Q.54** **NAT** **[2 Marks]**

Let $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 < 1\}$. Then $\frac{1}{\pi} \iiint_S ((x - 2y + z)^2 + (2x - y - z)^2 + (x - y + 2z)^2) dx dy dz =$ _____ (rounded off to two decimal places).

JAM 2024 **Q.57** **NAT** **[2 Marks]**

Let $c > 0$ be such that $\int_0^c e^{s^2} ds = 3$. Then $\int_0^c \left(\int_x^c e^{x^2+y^2} dy \right) dx =$ _____ (rounded off to one decimal place).

JAM 2023 **Q.7** **MCQ** **[1 Mark]**

$$\int_0^1 \int_0^{1-x} \cos(x^3 + y^2) dy dx - \int_0^1 \int_0^{1-y} \cos(x^3 + y^2) dx dy =:$$

- (A) 0
 (B) $\frac{\cos 1}{2}$
 (C) $\frac{\sin 1}{2}$
 (D) $\cos(\frac{1}{2}) - \sin(\frac{1}{2})$

JAM 2023

Q.18

MCQ

[2 Marks]

The curved surface area of $S = \{(x, y, z) \in \mathbb{R}^3 : z^2 = (x-1)^2 + (y-2)^2\}$ between $z = 2$ and $z = 3$ is:

- (A) $4\pi\sqrt{2}$
 (B) $5\pi\sqrt{2}$
 (C) 9π
 (D) $9\pi\sqrt{2}$

JAM 2023

Q.26

MCQ

[2 Marks]

Let $f(x, y) = \iint_{(u-x)^2 + (v-y)^2 \leq 1} e^{-\sqrt{(u-x)^2 + (v-y)^2}} du dv$. Then $\lim_{n \rightarrow \infty} f(n, n^2)$ is:

- (A) non-existent
 (B) 0
 (C) $\pi(1 - e^{-1})$
 (D) $2\pi(1 - 2e^{-1})$

JAM 2023

Q.31

MSQ

[2 Marks]

For $t \in (0, 1)$, let $P_t = \{(x, y, z) : (x^2 + y^2)z = 1, t^2 \leq x^2 + y^2 \leq 1\}$ and a_t be its surface area. Then:

- (A) $a_t = \iint_{t^2 \leq x^2 + y^2 \leq 1} \sqrt{1 + \frac{4x^2}{(x^2 + y^2)^4} + \frac{4y^2}{(x^2 + y^2)^4}} dx dy$
 (B) $a_t = \iint_{t^2 \leq x^2 + y^2 \leq 1} \sqrt{1 + \frac{4x^2}{(x^2 + y^2)^2} + \frac{4y^2}{(x^2 + y^2)^2}} dx dy$
 (C) $\lim_{t \rightarrow 0^+} a_t$ does NOT exist
 (D) $\lim_{t \rightarrow 0^+} a_t$ exists

JAM 2023

Q.48

NAT

[1 Mark]

Let V be the volume of $S = \{(x, y, z) : xy \leq z \leq 4, x^2 + y^2 \leq 1\}$. Then $\frac{V}{\pi} = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2023

Q.55

NAT

[2 Marks]

Let S be the triangular region with vertices $(0, 0)$, $(0, \frac{\pi}{2})$ and $(\frac{\pi}{2}, 0)$. Then $\iint_S \sin x \cos y dx dy = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2022

Q.11

MCQ

[2 Marks]

Let $G = \{(s, t) : 0 < s < 1, 0 < t < 1\}$ and $T(s, t) = (\frac{\pi s(1-t)}{2}, \frac{\pi(1-s)}{2})$. The area of $T(G)$ is:

- (A) $\frac{\pi}{4}$

- (B) $\frac{\pi^2}{4}$
 (C) $\frac{\pi^2}{8}$
 (D) 1

JAM 2022 **Q.48** **NAT** [1 Mark]

Consider the region $G = \{(x, y, z) \in \mathbb{R}^3 : 0 < z < x^2 - y^2, x^2 + y^2 < 1\}$. Then the volume of G is equal to _____ (rounded off to two decimal places).

JAM 2022 **Q.52** **NAT** [2 Marks]

If $G = \{(x, y) : x^2 + y^2 < 1, \frac{x}{\sqrt{3}} < y < \sqrt{3}x, x > 0, y > 0\}$, then $\frac{200}{\pi} \iint_G x^2 dx dy =$ _____ (rounded off to two decimal places).

2.3 Differential Equations

JAM 2026 **Q.4** **MCQ** [1 Mark]

The general solution of $x \sin(\frac{y}{x}) \frac{dy}{dx} = y \sin(\frac{y}{x}) + \frac{x}{2}$ ($x > 0$) is:

- (A) $\cos(\frac{x}{y}) + \log_e(x^2) = K$
 (B) $\cos(\frac{y}{x}) + \log_e(x^2) = K$
 (C) $\cos(\frac{x}{y}) + \log_e(\sqrt{x}) = K$
 (D) $\cos(\frac{y}{x}) + \log_e(\sqrt{x}) = K$

JAM 2026 **Q.16** **MCQ** [2 Marks]

The orthogonal trajectories of the family $y = -3x - 3 + me^x$ are given by:

- (A) $x = \frac{y}{3} - \frac{1}{9} + ke^{-3y}$
 (B) $x = \frac{y}{3} - \frac{1}{9} + ke^{3y}$
 (C) $x = -\frac{y}{3} + \frac{1}{9} + ke^{-3y}$
 (D) $x = -\frac{y}{3} + \frac{1}{9} + ke^{3y}$

JAM 2026 **Q.18** **MCQ** [2 Marks]

Let $\phi : (-\frac{\pi}{2}, \frac{\pi}{2}) \rightarrow \mathbb{R}$ solve $(\cos x) \frac{dy}{dx} - y = 2y^2(\sin x - 1) \cos x$ with $\phi(0) = \frac{1}{2}$. Then $\phi(\frac{\pi}{4}) =$:

- (A) 2
 (B) $\frac{3+2\sqrt{2}}{2}$
 (C) $\frac{1}{2}$
 (D) $\frac{1}{\sqrt{2}}$

JAM 2026 **Q.23** **MCQ** [2 Marks]

A particular solution of $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = \sin(e^x)$ is:

- (A) $-e^{-x} \sin(e^x)$

- (B) $-e^{-2x} \sin(e^x)$
 (C) $-e^{-x}(\sin(e^x) + \cos(e^x))$
 (D) $-e^{-2x}(\sin(e^x) + \cos(e^x))$

JAM 2026

Q.35

MSQ

[2 Marks]

Integrating factor(s) for $(2y \cos x - xy \sin x)dx + 2x \cos x dy = 0$ on $x \in (0, \frac{\pi}{4})$ is/are:

- (A) $\frac{1}{xy}$
 (B) xy
 (C) $\sec x$
 (D) $\sqrt{\sec x}$

JAM 2026

Q.45

NAT

[1 Mark]

If $(y^3 + \alpha xy^4 - 5x + \cos 2y)dx + (3xy^2 + 20x^2y^3 + \beta x \sin 2y)dy = 0$ is exact, then $\alpha + \beta = \underline{\hspace{2cm}}$.

JAM 2026

Q.51

NAT

[2 Marks]

If $y(x) = C_1 e^{-x} + C_2 e^{2x} + \alpha x e^{-x}$ is the general solution of $\frac{d^2 y}{dx^2} + \beta \frac{dy}{dx} + \gamma y = -e^{-x}$, then $\alpha(\beta + \gamma) = \underline{\hspace{2cm}}$.

JAM 2025

Q.2

MCQ

[1 Mark]

For which choice of $N(x, y)$ is $(e^x \sin y - 2y \sin x)dx + N(x, y)dy = 0$ exact?

- (A) $e^x \sin y + 2 \cos x$
 (B) $e^x \cos y + 2 \cos x$
 (C) $e^x \cos y + 2 \sin x$
 (D) $e^x \sin y + 2 \sin x$

JAM 2025

Q.5

MCQ

[1 Mark]

General solution of $\frac{d^2 y}{dx^2} - 8 \frac{dy}{dx} + 16y = 2e^{4x}$ is:

- (A) $\alpha_1 e^{4x} + \alpha_2 x e^{4x} + x^2 e^{4x}$
 (B) $\alpha_1 e^{4x} + \alpha_2 x e^{4x} + 2x^2 e^{4x}$
 (C) $\alpha_1 e^{-4x} + \alpha_2 e^{4x} + 2x^2 e^{4x}$
 (D) $\alpha_1 x e^{-4x} + \alpha_2 x^2 e^{-4x} + x^2 e^{4x}$

JAM 2025

Q.11

MCQ

[2 Marks]

Orthogonal trajectory to $yx^2 = \lambda$ passing through $(2, 1)$ is:

- (A) $y = \frac{x^2}{4}$
 (B) $x^2 - 2y^2 = 2$
 (C) $x - y = 1$
 (D) $2x - y^2 = 3$

JAM 2025**Q.12****MCQ****[2 Marks]**

Let ϕ solve $x \frac{dy}{dx} = (\ln y - \ln x)y$ with $\phi(1) = e^2$. Then $\phi(2) =$:

- (A) e^2
- (B) $2e^3$
- (C) $3e^2$
- (D) $6e^3$

JAM 2025**Q.14****MCQ****[2 Marks]**

Let ϕ solve $\frac{dy}{dx} = (y-1)(y-3)$ with $\phi(0) = 2$. Then:

- (A) $\lim_{x \rightarrow \infty} \phi(x) = 0$
- (B) $\lim_{x \rightarrow \ln \sqrt{2}} \phi(x) = 1$
- (C) $\lim_{x \rightarrow -\infty} \phi(x) = 3$
- (D) $\lim_{x \rightarrow \ln(1/\sqrt{2})} \phi(x) = 6$

JAM 2025**Q.29****MCQ****[2 Marks]**

Let $\phi(x) = \int_0^x \phi(t)dt + \sin x$. Then $\lim_{x \rightarrow \pi/2} (\phi(x) - e^x) =$:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

JAM 2025**Q.32****MSQ****[2 Marks]**

If $\mu Mdx + \mu Ndy = dw$, which of the following is/are TRUE?

- (A) μw is an integrating factor for $Mdx + Ndy = 0$
- (B) μw^2 is an integrating factor for $Mdx + Ndy = 0$
- (C) $w(x, y) = w(0, 0) + \int_0^x (\mu M)(s, 0)ds + \int_0^y (\mu N)(x, t)dt$
- (D) $w(x, y) = w(0, 0) + \int_0^x (\mu M)(s, y)ds + \int_0^x (\mu N)(0, t)dt$

JAM 2025**Q.33****MSQ****[2 Marks]**

Let $\phi : (-1, \infty) \rightarrow (0, \infty)$ solve $\frac{dy}{dx} - 2ye^x = 2e^x \sqrt{y}$ with $\phi(0) = 1$. Then:

- (A) ϕ is unbounded
- (B) $\lim_{x \rightarrow \ln 2} \phi(x) = (2e - 1)^2$
- (C) $\lim_{x \rightarrow \ln 2} \phi(x) = \sqrt{2e - 1}$
- (D) ϕ is strictly increasing on $(0, \infty)$

JAM 2025**Q.44****NAT****[1 Mark]**

Let ϕ solve $4y'' + 16y' + 25y = 0$ with $\phi(0) = 1, \phi'(0) = -\frac{1}{2}$. Then $\lim_{x \rightarrow \pi/6} e^{2x} \phi(x) =$ _____ (rounded off to two decimal places).

JAM 2025**Q.55****NAT****[2 Marks]**

Let ϕ solve $x^2 y'' - xy' + y = 6x \ln x$ on $(0, \infty)$ with $\phi(1) = -3, \phi(e) = 0$. Then $|\phi'(1)| =$ _____ (rounded off to two decimal places).

JAM 2025**Q.56****NAT****[2 Marks]**

Let ϕ solve $\frac{dy}{dx} + 2xy = 2 + 4x^2$ with $\phi(0) = 0$. Then $\phi(2) =$ _____ (rounded off to two decimal places).

JAM 2024**Q.1****MCQ****[1 Mark]**

Let y_c solve $\frac{dy}{dx} - y + y^3 = 0, y(0) = c > 0$. Then for every $c > 0$:

- (A) $\lim_{x \rightarrow \infty} y_c(x) = 0$
- (B) $\lim_{x \rightarrow \infty} y_c(x) = 1$
- (C) $\lim_{x \rightarrow \infty} y_c(x) = e$
- (D) limit does not exist

JAM 2024**Q.3****MCQ****[1 Mark]**

Let $y(x)$ solve $\frac{dy}{dx} = 1 + y \sec x$ on $(-\pi/2, \pi/2)$ with $y(0) = 0$. Then $y(\pi/6) =$:

- (A) $\sqrt{3} \log(\frac{3}{2})$
- (B) $(\frac{\sqrt{3}}{2}) \log(\frac{3}{2})$
- (C) $(\frac{\sqrt{3}}{2}) \log 3$
- (D) $\sqrt{3} \log 3$

JAM 2024**Q.4****MCQ****[1 Mark]**

Differential equation for orthogonal trajectories of $x^2 + 2hxy + y^2 = 1$ is:

- (A) $(x^2 y - y^3 + y) \frac{dy}{dx} - (xy^2 - x^3 + x) = 0$
- (B) $(x^2 y - y^3 + y) \frac{dy}{dx} + (xy^2 - x^3 + x) = 0$
- (C) $(x^2 y + y^3 + y) \frac{dy}{dx} - (xy^2 + x^3 + x) = 0$
- (D) $(x^2 y + y^3 + y) \frac{dy}{dx} + (xy^2 + x^3 + x) = 0$

JAM 2024**Q.10****MCQ****[1 Mark]**

Solution of $y'' + y = g(x)$ with $y(0) = 0, y'(0) = 1$ is:

- (A) $y(x) = \sin x - \int_0^x \sin(x-t)g(t)dt$
- (B) $y(x) = \sin x + \int_0^x \sin(x-t)g(t)dt$
- (C) $y(x) = \sin x - \int_0^x \cos(x-t)g(t)dt$
- (D) $y(x) = \sin x + \int_0^x \cos(x-t)g(t)dt$

JAM 2024**Q.18****MCQ****[2 Marks]**

For $y(8x - 9y)dx + 2x(x - 3y)dy = 0$:

- (A) Not exact, has x^2 as integrating factor
- (B) Exact and homogeneous
- (C) Not exact, does not have x^2 as I.F.
- (D) Not homogeneous, has x^2 as I.F.

JAM 2024**Q.23****MCQ****[2 Marks]**

Let y_α solve $\frac{dy}{dx} + 2y = \frac{1}{1+x^2}$ with $y(0) = \alpha$. Then:

- (A) $\lim_{x \rightarrow \infty} y_\alpha(x) = 0$ for every $\alpha \in \mathbb{R}$
- (B) $\lim_{x \rightarrow \infty} y_\alpha(x) = 1$ for every $\alpha \in \mathbb{R}$
- (C) $\exists \alpha$ such that limit exists and $\neq 0, 1$
- (D) $\exists \alpha$ for which limit does not exist

JAM 2024**Q.26****MCQ****[2 Marks]**

Let $y'' - 2y' + y = 2e^x$. P: $f > 0 \implies f' > 0$. Q: $f' > 0 \implies f > 0$.

- (A) P true, Q false
- (B) P false, Q true
- (C) Both true
- (D) Both false

JAM 2024**Q.42****NAT****[1 Mark]**

Let $y : \mathbb{R} \rightarrow \mathbb{R}$ be the solution to the differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 1$ satisfying $y(0) = 0$ and $y'(0) = 1$. Then, $\lim_{x \rightarrow \infty} y(x)$ equals _____ (rounded off to two decimal places).

JAM 2024**Q.43****NAT****[1 Mark]**

Let y_α solve $2y'' - y' - y = 0, y(0) = 1, y'(0) = \alpha > 0$. Smallest α for which y_α has no critical points in $\mathbb{R}$ is _____.

JAM 2024**Q.51****NAT****[2 Marks]**

Number of $\alpha \in (-2\pi, 0) \cap \mathbb{Z}$ for which all real solutions of $x^2y'' + \alpha xy' + y = 0$ approach 0 as $x \rightarrow 0^+$ is _____.

JAM 2023**Q.4****MCQ****[1 Mark]**

Consider the initial value problem $\frac{dy}{dx} + \alpha y = 0, y(0) = 1$, where $\alpha \in \mathbb{R}$. Then:

- (A) there is an α such that $y(1) = 0$
- (B) there is a unique α such that $\lim_{x \rightarrow \infty} y(x) = 0$
- (C) there is NO α such that $y(2) = 1$
- (D) there is a unique α such that $y(1) = 2$

JAM 2023**Q.22****MCQ****[2 Marks]**

Orthogonal trajectory to $x^2 + y^2 = 2x + 4y + k$ passing through $(2, 3)$ also passes through:

- (A) $(3, 4)$
- (B) $(-1, 1)$
- (C) $(1, 0)$
- (D) $(3, 5)$

JAM 2023**Q.33****MSQ****[2 Marks]**

Let $y : (\sqrt{2/3}, \infty) \rightarrow \mathbb{R}$ solve $(2x - y)y' + (2y - x) = 0$ with $y(1) = 3$. Then:

- (A) $y(3) = 1$
- (B) $y(2) = 4 + \sqrt{10}$
- (C) y' bounded on $(\sqrt{2/3}, 1)$
- (D) y' bounded on $(1, \infty)$

JAM 2023**Q.43****NAT****[1 Mark]**

If $y'' - 2y' + y = e^x$, $y(0) = 0$, $y'(0) = -1/2$, then $y(1) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2023**Q.53****NAT****[2 Marks]**

Let $y : (1, \infty) \rightarrow \mathbb{R}$ solve $y'' - \frac{2y}{(1-x)^2} = 0$ with $y(2) = 1$ and $\lim_{x \rightarrow \infty} y(x) = 0$. Then $y(3) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2023**Q.59****NAT****[2 Marks]**

Let $S = \{\alpha \in \mathbb{R} : \text{solution of } y' = y(2 - y), y(0) = \alpha \text{ exists on } [0, \infty)\}$. Then $\min S = \underline{\hspace{2cm}}$.

JAM 2022**Q.13****MCQ****[2 Marks]**

Let $u''(x) = \frac{u(x)}{1+x^2}$ on $\mathbb{R}$ with $u(0) > 0$, $u'(0) > 0$. I: uu' is increasing on $[0, \infty)$. II: u is increasing on $[0, \infty)$.

- (A) Both false
- (B) Both true
- (C) I false, II true
- (D) I true, II false

JAM 2022**Q.21****MCQ****[2 Marks]**

On $(-c, c)$, let $y(x)$ solve $\frac{dy}{dx} = y^2 - 1 + \cos x$, $y(0) = 0$. Then:

- (A) Local max at origin
- (B) Local min at origin
- (C) Strictly increasing on $(-\delta, \delta)$
- (D) Strictly decreasing on $(-\delta, \delta)$

JAM 2022**Q.23****MCQ****[2 Marks]**

Consider $y'' + ay' + y = \sin x$.

- (A) If $a = 0$, all solutions unbounded over $\mathbb{R}$
- (B) If $a = 1$, all solutions unbounded over $(0, \infty)$
- (C) If $a = 1$, all solutions tend to 0 as $x \rightarrow \infty$
- (D) If $a = 2$, all solutions bounded over $(-\infty, 0)$

JAM 2022**Q.25****MCQ****[2 Marks]**

Let ϕ solve $(x^2 + y^2)dx - 4xy dy = 0$ with $\phi(1) = 1$. Then:

- (A) $(3\phi^2 + x^2)^2 = 4x$
- (B) $(3\phi^2 - x^2)^2 = 4x$
- (C) $(3\phi^2 + x^2)^2 = 4\phi$
- (D) $(3\phi^2 - x^2)^2 = 4\phi$

JAM 2022**Q.31****MSQ****[2 Marks]**

Let $y' = x^2 + y^2 + 1$ with $y(0) = 0$ on $(-c, c)$. Then:

- (A) $y(x)$ is odd
- (B) $y(x)$ is even
- (C) $(y(x))^2$ has local min at 0
- (D) $(y(x))^2$ has local max at 0

JAM 2022**Q.35****MSQ****[2 Marks]**

Consider $y'' + 4y = \sin(ax), x \in \mathbb{R}$.

- (A) All solutions periodic for every a
- (B) All solutions periodic for $a \in \mathbb{R} \setminus \{-2, 2\}$
- (C) All solutions periodic for $a \in \mathbb{Q} \setminus \{-2, 2\}$
- (D) If $a \in \mathbb{R} \setminus \mathbb{Q}$, then unique periodic solution

JAM 2022**Q.43****NAT****[1 Mark]**

Let $y' + 3x^2y = x^2, y(0) = 4$. Then $\lim_{x \rightarrow \infty} y(x) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2022**Q.47****NAT****[1 Mark]**

Let $xy^2y' + y^3 = \frac{\sin x}{x}, y(\frac{\pi}{2}) = 0$. Then $y(\frac{5\pi}{2}) = \underline{\hspace{2cm}}$ (rounded off to two decimal places).

JAM 2022**Q.49****NAT****[1 Mark]**

Let $x^2y'' + xy' - 4y = x^2$ on $(0, \infty)$ such that $\lim_{x \rightarrow 0^+} y(x)$ exists and $y(1) = 1$. Then $y'(1) = \underline{\hspace{2cm}}$.

JAM 2022**Q.50****NAT****[1 Mark]**

Orthogonal trajectory to $y = \frac{c(1-\cos x)}{\sin x}$ passing through $(\frac{\pi}{3}, 1)$. If $(\frac{\pi}{4}, a)$ lies on it, then $a^4 = \underline{\hspace{2cm}}$.

3 Linear Algebra and Algebra

3.1 Basic Algebra

JAM 2026

Q.38

MSQ

[2 Marks]

Let $\binom{n}{r}$ denote the number of ways of choosing r distinct objects out of n distinct objects. Which of the following statements is/are TRUE?

- (A) $\sum_{k=0}^6 (-1)^k \binom{13}{2k} = -64$
- (B) $\sum_{k=0}^6 (-1)^k \binom{13}{2k+1} = 128$
- (C) $\sum_{k=0}^6 (-1)^k \binom{13}{2k} = 64$
- (D) $\sum_{k=0}^6 (-1)^k \binom{13}{2k+1} = -128$

JAM 2026

Q.58

NAT

[2 Marks]

A fruit shop has 4 different types of bananas. The number of ways in which 12 bananas can be bought with at least one banana from each type, is _____.

JAM 2026

Q.60

NAT

[2 Marks]

$$\binom{5}{0} + \binom{6}{1} + \binom{7}{2} + \binom{8}{3} + \binom{9}{4} + \binom{10}{5} + \binom{11}{6} = \text{_____}.$$

3.2 Matrices and Systems of Linear Equations

JAM 2026

Q.8

MCQ

[1 Mark]

Let A be an invertible 5×5 real matrix. B is obtained by swapping columns 2 and 3 of A , then adding 3 times column 3 to column 5. Then B^{-1} is obtained by:

- (A) swapping rows 2 and 3 of A^{-1} , then adding 3 times row 3 to row 5
- (B) swapping rows 2 and 3 of A^{-1} , then adding -3 times row 5 to row 3
- (C) adding -3 times row 3 to row 5, then swapping rows 2 and 3 of A^{-1}
- (D) adding 3 times row 5 to row 3, then swapping rows 2 and 3 of A^{-1}

JAM 2026

Q.14

MCQ

[2 Marks]

Let A be a 5×3 real matrix of rank 2 and b be a non-zero 5×1 real column vector. Suppose $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$ are two solutions of the system of linear equations $Ax = b$. Which ONE of the following is also a solution of $Ax = b$?

- (A) $\begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$
- (B) $\begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix}$

(C) $\begin{pmatrix} 5 \\ 6 \\ 9 \end{pmatrix}$

(D) $\begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$

JAM 2026

Q.25

MCQ

[2 Marks]

Let P be a nonzero $n \times n$ complex matrix such that $v^* P v \geq 0$ for all $v \in \mathbb{C}^n$. Then:

- (A) All eigenvalues of P are negative
- (B) If $Pv_1 = v_2$ and $Pv_2 = v_1$, then $v_1 = v_2$
- (C) $v^* P v \neq 0$ for all nonzero v
- (D) All eigenvalues of P^* are negative

JAM 2026

Q.37

MSQ

[2 Marks]

Let $P = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \in M_5(\mathbb{C})$. Which of the following is/are TRUE?

- (A) $\text{nullity}(P - I) \geq 2$
- (B) P has 4 distinct eigenvalues in $\mathbb{C}$
- (C) P has 3 distinct eigenvalues in $\mathbb{R}$
- (D) If λ is an eigenvalue, $\exists n \in \mathbb{N}$ such that $\lambda^n = 1$

JAM 2026

Q.39

MSQ

[2 Marks]

Let $P \in M_{4,6}(\mathbb{R})$ and $Q \in M_{6,4}(\mathbb{R})$ such that $PQ = 0$ and $QP = 0$. Then:

- (A) $\text{range}(P) \subseteq \text{null}(Q)$ and $\text{range}(Q) \subseteq \text{null}(P)$
- (B) $\text{rank}(P) + \text{rank}(Q) \leq 4$
- (C) If $\text{range}(P) = \text{null}(Q)$, then $\text{rank}(P) + \text{rank}(Q) = 4$
- (D) $\text{range}(Q) = \text{null}(P)$

JAM 2026

Q.49

NAT

[1 Mark]

Let $A \in M_3(\mathbb{C})$. Suppose $v = \begin{pmatrix} \sqrt{5}i \\ 2i \\ x \end{pmatrix} \in \text{null}(A) \cap \text{range}(A^T)$. Then $|x| = \underline{\hspace{2cm}}$.

JAM 2026

Q.50

NAT

[1 Mark]

Let P be a 5×5 real matrix with $\det(P) = 2$. Let Q be the matrix of cofactors of P . Then $\det(Q) = \underline{\hspace{2cm}}$.

JAM 2026

Q.56

NAT

[2 Marks]

Let $M \in M_3(\mathbb{R})$ satisfy $M \begin{pmatrix} \sqrt{15} \\ \sqrt{15} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3} \\ \sqrt{2} \\ -5 \end{pmatrix}$, $M \begin{pmatrix} 0 \\ 0 \\ \sqrt{30} \end{pmatrix} = \begin{pmatrix} \sqrt{15} \\ \sqrt{10} \\ \sqrt{5} \end{pmatrix}$, $M \begin{pmatrix} -\sqrt{15} \\ \sqrt{15} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{12} \\ -\sqrt{18} \\ 0 \end{pmatrix}$.

Then $\det(M) =$ _____.

JAM 2026

Q.57

NAT

[2 Marks]

If $x + 2y + 2z = 1$, $2x + 3y + z = 2$, $ax + 5y + bz = b$ has infinitely many solutions, then $a + b =$

_____.

JAM 2026

Q.59

NAT

[2 Marks]

Let $A \in M_3(\mathbb{R})$ represent reflection about $\{(a, b, -a - b) : a, b \in \mathbb{R}\}$. Sum of diagonal elements of A is _____.

JAM 2025

Q.7

MCQ

[1 Mark]

Let $M = \begin{pmatrix} 2 & 0 & -1 \\ 4 & 1 & -4 \\ 2 & 0 & x \end{pmatrix}$. If 0 is an eigenvalue of M , then $(M^4 + M) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ equals:

(A) $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

(B) $\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix}$

(C) $\begin{pmatrix} 5 \\ 0 \\ 5 \end{pmatrix}$

(D) $\begin{pmatrix} 17 \\ 0 \\ 17 \end{pmatrix}$

JAM 2025

Q.15

MCQ

[2 Marks]

Let $M = \begin{pmatrix} 6 & 2 & -6 & 8 \\ 5 & 3 & -9 & 8 \\ 3 & 1 & -2 & 4 \end{pmatrix}$ and system $6x_1 + 2x_2 - 6x_3 + 8x_4 = 8$, $5x_1 + 3x_2 - 9x_3 + 8x_4 = 16$, $3x_1 + x_2 - 2x_3 + 4x_4 = 32$.

(A) $\text{rank}(M) = 3$, system has solution

(B) $\text{rank}(M) = 3$, no solution

(C) $\text{rank}(M) = 2$, has solution

(D) $\text{rank}(M) = 2$, no solution

JAM 2025

Q.16

MCQ

[2 Marks]

Let $M = \begin{pmatrix} -2 & 0 & 0 \\ 3 & 2 & 3 \\ 4 & -1 & x \end{pmatrix}$ have eigenvalues $-2, 3$. If $M^3 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 125 \\ 125 \end{pmatrix}$, then:

(A) $x = 5$, $M^2 + M$ invertible

(B) $x \neq 5$, $M^2 + M$ invertible

(C) $x = 5$, $M^2 + M$ not invertible

(D) $x \neq 5$, $M^2 + M$ not invertible

JAM 2025

Q.22

MCQ

[2 Marks]

Let $M = (m_{ij})_{3 \times 3}$ be invertible, $\sigma = (123) \in S_3$, and $M_\sigma = (m_{i\sigma(j)})$. Then:

(A) $\det(M) = \det(M_\sigma)$, $\text{nullity}(M - M_\sigma) = 0$

(B) $\det(M) = -\det(M_\sigma)$, $\text{nullity}(M - M_\sigma) = 1$

(C) $\det(M) = \det(M_\sigma)$, $\text{nullity}(M - M_\sigma) = 1$

(D) $\det(M) = -\det(M_\sigma)$, $\text{nullity}(M - M_\sigma) = 0$

JAM 2024

Q.9

MCQ

[1 Mark]

P: Unique solution to $Ax = b$ ($m \times n$) $\implies m = n$. Q: $u, v \in V \setminus W \implies u + v \in V \setminus W$.

(A) Both true

(B) P true, Q false

(C) P false, Q true

(D) Both false

JAM 2024

Q.15

MCQ

[2 Marks]

Let $a = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} \\ 0 \end{pmatrix}$. Consider the following two statements.

P: The matrix $I_4 - aa^T$ is invertible.

Q: The matrix $I_4 - 2aa^T$ is invertible.

Then, which one of the following holds?

(A) P is false but Q is true

(B) P is true but Q is false

(C) Both P and Q are true

(D) Both P and Q are false

JAM 2024

Q.16

MCQ

[2 Marks]

$A_{6 \times 5}, B_{5 \times 4}$. P: $\exists Z \neq 0$ s.t. $AZB = 0$. Q: $\exists Y \neq 0$ s.t. $BYA = 0$.

(A) Both true

(B) P true, Q false

(C) P false, Q true

(D) Both false

JAM 2024**Q.37****MSQ****[2 Marks]**

For $A, B, C \in M_{10}(\mathbb{R})$ with $A = BC$:

- (A) $\text{Rowspace}(A) \subseteq \text{Rowspace}(B)$
- (B) $\text{Rowspace}(A) \subseteq \text{Rowspace}(C)$
- (C) $\text{Colspace}(A) \subseteq \text{Colspace}(B)$
- (D) $\text{Colspace}(A) \subseteq \text{Colspace}(C)$

JAM 2024**Q.44****NAT****[1 Mark]**

Let $M = \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix}$. The $(4, 1)$ -entry of M^{-1} is _____ (rounded off to two decimal places).

JAM 2024**Q.50****NAT****[1 Mark]**

Let $M = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 \\ 2 & 0 & 0 & 0 & -4 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 3 \\ 0 & 0 & 0 & 2 & 2 \end{pmatrix}$. If $p(x)$ is char. poly, then $p(2) - 1 = \underline{\hspace{2cm}}$.

JAM 2023**Q.2****MCQ****[1 Mark]**

Let $v_1, \dots, v_9$ be columns of 9×9 matrix A with $\sum a_i v_i = 0$ (not all $a_i = 0$). Then $Ax = \sum v_i$ has:

- (A) no solution
- (B) a unique solution
- (C) more than one but only finitely many solutions
- (D) infinitely many solutions

JAM 2023**Q.13****MCQ****[2 Marks]**

Let $A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ -2 & 2 & 2 \end{pmatrix}$ and $B = A^5 + A^4 + I_3$. Which is NOT an eigenvalue of B ?

- (A) 1
- (B) 2
- (C) 49
- (D) 3

JAM 2023**Q.14****MCQ****[2 Marks]**

The system of linear equations in x_1, x_2, x_3 : $\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & 1 \\ 2 & 3 & \alpha \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ \beta \end{pmatrix}$, where $\alpha, \beta \in \mathbb{R}$, has:

- (A) at least one solution for any α and β
- (B) a unique solution for any β when $\alpha \neq 1$

- (C) NO solution for any α when $\beta \neq 5$
 (D) infinitely many solutions for any α when $\beta = 5$

JAM 2023 **Q.56** **NAT** **[2 Marks]**

Let $A = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 4 & 4 & 4 \end{pmatrix}$ and $AB = 0$ for 5×5 matrix B . Max rank of B is _____.

JAM 2023 **Q.58** **NAT** **[2 Marks]**

Max number of linearly independent eigenvectors of $\begin{pmatrix} 1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 1 & 3 \end{pmatrix}$ is _____.

JAM 2022 **Q.1** **MCQ** **[1 Mark]**

Let $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. If $M^8 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$, then $x =$:

- (A) 21
 (B) 22
 (C) 34
 (D) 35

JAM 2022 **Q.2** **MCQ** **[1 Mark]**

Rank of $\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$ is:

- (A) 1
 (B) 2
 (C) 3
 (D) 4

JAM 2022 **Q.16** **MCQ** **[2 Marks]**

Let $P \in M_4(\mathbb{R})$ with $P^4 = 0, P^3 \neq 0$. Which ONE is FALSE?

- (A) $\forall v \neq 0, \{v, Pv, P^2v, P^3v\}$ is linearly independent.
 (B) $\text{rank}(P^k) = 4 - k$ for $k = 1, 2, 3, 4$.
 (C) 0 is an eigenvalue of P .
 (D) If $Q^4 = 0, Q^3 \neq 0$, then $\exists S$ nonsingular such that $S^{-1}QS = P$.

JAM 2022 **Q.17** **MCQ** **[2 Marks]**

For $(X, Y) = XY - YX$. P: $(X, (Y, Z)) + (Y, (Z, X)) + (Z, (X, Y)) = 0$. Q: $(X, (Y, Z)) = ((X, Y), Z)$.

- (A) Both true

- (B) P true, Q false
 (C) P false, Q true
 (D) Both false

JAM 2022

Q.18

MCQ

[2 Marks]

Consider the system of linear equations: $x + y + t = 4$, $2x - 4t = 7$, $x + y + z = 5$, $x - 3y - z - 10t = \lambda$, where x, y, z, t are variables and λ is a constant. Then which one of the following is true?

- (A) If $\lambda = 1$, then the system has a unique solution.
 (B) If $\lambda = 2$, then the system has infinitely many solutions.
 (C) If $\lambda = 1$, then the system has infinitely many solutions.
 (D) If $\lambda = 2$, then the system has a unique solution.

JAM 2022

Q.30

MCQ

[2 Marks]

Let P be a 3×3 real matrix having eigenvalues $\lambda_1 = 0$, $\lambda_2 = 1$ and $\lambda_3 = -1$. Further, $v_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $v_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ are eigenvectors of P corresponding to λ_1, λ_2 and λ_3 , respectively. Then the entry in the first row and the third column of P is:

- (A) 0
 (B) 1
 (C) -1
 (D) 2

JAM 2022

Q.53

NAT

[2 Marks]

Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{pmatrix}$, $\|u\| = 1$, $\|v\| = 1$. If $Au = \sqrt{2}v$ and $A^T v = \sqrt{2}u$, then $|u_1 + 2\sqrt{2}v_1| = \underline{\hspace{2cm}}$.

JAM 2022

Q.57

NAT

[2 Marks]

$\det \begin{pmatrix} 11 & 10 & 10 & 10 \\ 10 & 11 & 10 & 10 \\ 10 & 10 & 11 & 10 \\ 10 & 10 & 10 & 11 \end{pmatrix} = \underline{\hspace{2cm}}.$

3.3 Finite Dimensional Vector Spaces

JAM 2026

Q.24

MCQ

[2 Marks]

Let $V = \text{span}\{e_1 + e_2, e_2 + e_3, e_3 + e_4 + e_5\} \subset \mathbb{R}^5$. Let $S = \{P \in M_{2,5}(\mathbb{R}) : \text{null}(P) = V\}$.

- (A) $S = \emptyset$
 (B) $S = \{0\}$
 (C) S is nonempty and S is not a subspace of $M_{2,5}(\mathbb{R})$
 (D) S is a nonzero subspace of $M_{2,5}(\mathbb{R})$

JAM 2026**Q.28****MCQ****[2 Marks]**

Let $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be linear preserving distances $d(F(P), F(Q)) = d(P, Q)$. Which ONE is FALSE?

- (A) F is an injective map.
- (B) F is a surjective map.
- (C) $\text{vect}(F(P), F(Q)) \cdot \text{vect}(F(R), F(S)) = \text{vect}(P, Q) \cdot \text{vect}(R, S)$
- (D) $\text{vect}(F(P), F(Q)) \times \text{vect}(F(R), F(S)) = \text{vect}(P, Q) \times \text{vect}(R, S)$

JAM 2026**Q.40****MSQ****[2 Marks]**

Let $V = \left\{ \frac{a+b\sqrt{2}}{c+d\sqrt{2}} : a, b, c, d \in \mathbb{Q}, c^2 + d^2 \neq 0 \right\}$. Which of the following is/are FALSE?

- (A) V is closed under the usual addition in $\mathbb{R}$.
- (B) V is a subspace of the real vector space $\mathbb{R}$.
- (C) V is a two dimensional subspace of the vector space $\mathbb{R}$ over $\mathbb{Q}$.
- (D) V is a four dimensional subspace of the vector space $\mathbb{R}$ over $\mathbb{Q}$.

JAM 2025**Q.6****MCQ****[1 Mark]**

Define $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by $T(x, y, z) = (x + z, 2x + 3y + 5z, 2y + 2z)$. Then:

- (A) T is one-one and T is NOT onto
- (B) T is NOT one-one and T is onto
- (C) T is one-one and T is onto
- (D) T is NOT one-one and T is NOT onto

JAM 2025**Q.8****MCQ****[1 Mark]**

Let $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be $T(p(x)) = p(x + 1)$ with matrix M w.r.t. $\{1, x, x^2\}$. Then:

- (A) The determinant of M is 2
- (B) The rank of M is 2
- (C) 1 is the only eigenvalue of M
- (D) The nullity of M is 2

JAM 2025**Q.24****MCQ****[2 Marks]**

Let $f_1, f_2, f_3 : \mathbb{R}^4 \rightarrow \mathbb{R}$ linear with $\ker(f_1) \subseteq \ker(f_2) \cap \ker(f_3)$. Let $T(v) = (f_1(v), f_2(v), f_3(v))$. Nullity of T is:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

JAM 2025**Q.35****MSQ****[2 Marks]**

For $u_1 = (1, 0, 0, -1), u_2 = (0, 2, 0, -1), u_3 = (0, 0, 1, -1), u_4 = (0, 0, 0, 1)$ in $\mathbb{R}^4$:

- (A) $\{u_1, u_2, u_3, u_4\}$ is a linearly independent set in $\mathbb{R}^4$
 (B) $\{u_1 - u_2, u_2 - u_3, u_3 - u_4, u_4 - u_1\}$ is NOT a linearly independent set in $\mathbb{R}^4$
 (C) $\{u_1, -u_2, u_3, -u_4\}$ is NOT a linearly independent set in $\mathbb{R}^4$
 (D) $\{u_1 + u_2, u_2 + u_3, u_3 + u_4, u_4 + u_1\}$ is a linearly independent set in $\mathbb{R}^4$

JAM 2025**Q.40****MSQ****[2 Marks]**

$V_1 = \text{span}\{(1, 2, 3), (1, 1, 0)\}, V_2 = \text{span}\{(1, -1, 0)\}, V_3 = \text{span}\{(1, 1, 1)\}, V_4 = \text{span}\{(1, 3, 6)\}, V_5 = \text{span}\{(1, 0, -3)\}.$

- (A) $V_1 \cup V_2$ is a subspace of $\mathbb{R}^3$
 (B) $V_1 \cup V_3$ is a subspace of $\mathbb{R}^3$
 (C) $V_1 \cup V_4$ is a subspace of $\mathbb{R}^3$
 (D) $V_1 \cup V_5$ is a subspace of $\mathbb{R}^3$

JAM 2025**Q.46****NAT****[1 Mark]**

$\dim(V_1 \cap V_2 \cap V_3)$ for $V_1 : x + y + 2w = 0, V_2 : 2y + z + w = 0, V_3 : x + 3y + z + 3w = 0$ in $\mathbb{R}^4$ is _____.

JAM 2025**Q.47****NAT****[1 Mark]**

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ linear with $T(1, 1, 1) = 0, T(1, -1, 1) = 0, T(0, 0, 1) = 16$. Then $T(\frac{1}{2}, \frac{2}{3}, \frac{3}{4}) =$ _____.

JAM 2025**Q.49****NAT****[1 Mark]**

$T(p(x)) = xp'(x)$ and $S(p(x)) = (x+1)p'(x)$ on $P_4(\mathbb{R})$. Nullity of $S \circ T$ is _____.

JAM 2025**Q.58****NAT****[2 Marks]**

Dimension of $H = \{p(x) \in P_3(\mathbb{R}) : xp'(x) = 3p(x)\}$ is _____.

JAM 2024**Q.13****MCQ****[2 Marks]**

Nonzero subspace $V \subseteq M_7(\mathbb{C})$ where every non-zero matrix is invertible. $\dim(V)$ over $\mathbb{C}$ is:

- (A) 1
 (B) 2
 (C) 7
 (D) 49

JAM 2024**Q.17****MCQ****[2 Marks]**

Let $P_{11}(x)$ be the real vector space of polynomials in x with real coefficients and having degree at most 11, together with the zero polynomial. Let $E = \{s_0(x), s_1(x), \dots, s_{11}(x)\}$ and $F = \{r_0(x), r_1(x), \dots, r_{11}(x)\}$ be subsets of $P_{11}(x)$ having 12 elements each and satisfying $s_0(3) = s_1(3) = \dots = s_{11}(3) = 0$ and $r_0(4) = r_1(4) = \dots = r_{11}(4) = 1$. Then, which one of the following is TRUE?

- (A) Any such E is not necessarily linearly dependent and any such F is not necessarily linearly dependent
- (B) Any such E is necessarily linearly dependent but any such F is not necessarily linearly dependent
- (C) Any such E is not necessarily linearly dependent but any such F is necessarily linearly dependent
- (D) Any such E is necessarily linearly dependent and any such F is necessarily linearly dependent

JAM 2024

Q.22

MCQ

[2 Marks]

$$T(f(x)) = f(x) + f'(x) \text{ on } P_7(x).$$

- (A) T is not a surjective linear transformation
- (B) There exists $k \in \mathbb{N}$ such that T^k is the zero linear transformation
- (C) 1 and 2 are the eigenvalues of T
- (D) There exists $r \in \mathbb{N}$ such that $(T - I)^r$ is the zero linear transformation

JAM 2024

Q.45

NAT

[1 Mark]

$$\dim(\{f \in P_{12}(x) : f(-x) = f(x) \text{ and } f(2024) = 0\}) = \underline{\hspace{2cm}}.$$

JAM 2024

Q.49

NAT

[1 Mark]

Let $P_7(x)$ be the real vector space of polynomials in x with degree at most 7, together with the zero polynomial. For $r = 1, 2, \dots, 7$, define $s_r(x) = x(x-1)\cdots(x-(r-1))$ and $s_0(x) = 1$. If $x^5 = \sum_{k=0}^7 \alpha_{5,k} s_k(x)$, where $\alpha_{5,k} \in \mathbb{R}$, then $\alpha_{5,2}$ equals _____ (rounded off to two decimal places).

JAM 2024

Q.58

NAT

[2 Marks]

Dimension of continuous piecewise linear functions on $[0, 1]$ with nodes $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$ is _____.

JAM 2023

Q.3

MCQ

[1 Mark]

Which is a subspace of $\mathbb{R}^3$?

- (A) $\{(x, y, z) : (y+z)^2 + (2x-3y)^2 = 0\}$
- (B) $\{(x, y, z) : y \in \mathbb{Q}\}$
- (C) $\{(x, y, z) : yz = 0\}$
- (D) $\{(x, y, z) : x + 2y - 3z + 1 = 0\}$

JAM 2023

Q.12

MCQ

[2 Marks]

I: Linear map on $\mathbb{R}^3$ with range = null. II: Linear map on $\mathbb{R}^2$ with range = null.

- (A) Both I and II are TRUE
- (B) I is TRUE but II is FALSE
- (C) II is TRUE but I is FALSE
- (D) Both I and II are FALSE

JAM 2023

Q.15

MCQ

[2 Marks]

I: $\text{span}(S) = \mathbb{R}^2 \implies \text{span}(S \cap W) = W$. II: $\text{span}(S \cup T) = \text{span}(S) \cup \text{span}(T)$.

- (A) Both I and II are TRUE
- (B) I is TRUE but II is FALSE
- (C) II is TRUE but I is FALSE
- (D) Both I and II are FALSE

JAM 2023

Q.38

MSQ

[2 Marks]

Which of the following is/are true?

- (A) Every linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps lines onto points or lines
- (B) Every surjective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps lines onto lines
- (C) Every bijective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps pairs of parallel lines to pairs of parallel lines
- (D) Every bijective linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ maps pairs of perpendicular lines to pairs of perpendicular lines

JAM 2023

Q.39

MSQ

[2 Marks]

Which of the following is/are linear transformations?

- (A) $T(x) = \sin x$
- (B) $T(A) = \text{trace}(A)$
- (C) $T(x, y) = x + y + 1$
- (D) $T(p(x)) = p(1)$

JAM 2023

Q.42

NAT

[1 Mark]

Rank of $T : P_2(\mathbb{R}) \rightarrow P_4(\mathbb{R})$ given by $T(p(x)) = p(x^2)$ is _____.

JAM 2023

Q.57

NAT

[2 Marks]

Dimension of matrices in $M_3(\mathbb{R})$ with each row sum and column sum equal to zero is _____.

JAM 2022

Q.3

MCQ

[1 Mark]

Which is a subspace of $P_6(\mathbb{R})$?

- (A) $\{f : f(1/2) \notin \mathbb{Q}\}$
- (B) $\{f : f(1/2) = 1\}$
- (C) $\{f : f(1/2) = f(1)\}$
- (D) $\{f : f'(1/2) = 1\}$

JAM 2022

Q.20

MCQ

[2 Marks]

Let $S = \{P \in M_5(\mathbb{R}) : p_{ij} = p_{rs} \text{ for } i + r = j + s\}$. Which is FALSE?

- (A) S is a subspace of symmetric matrices
- (B) $\dim(S) = 5$
- (C) $\dim(S) = 11$

(D) 5 divides diagonal sum if entries are integers

JAM 2022

Q.26

MCQ

[2 Marks]

Dimension of $H = \{M \in M_4(\mathbb{C}) : M^T = \overline{M}\}$ over $\mathbb{R}$ is:

- (A) 6
- (B) 16
- (C) 15
- (D) 12

JAM 2022

Q.39

MSQ

[2 Marks]

Let $T : P_5 \rightarrow \mathbb{R}$ linear with $T(1) = 1$ and $T(x(x-1)\cdots(x-k+1)) = 1$ for $1 \leq k \leq 5$.

- (A) $T(x^4) = 15$
- (B) $T(x^3) = 5$
- (C) $T(x^4) = 14$
- (D) $T(x^3) = 3$

JAM 2022

Q.40

MSQ

[2 Marks]

Let P be a fixed 3×3 matrix with entries in $\mathbb{R}$. Which of the following maps from $M_3(\mathbb{R})$ to $M_3(\mathbb{R})$ is/are linear?

- (A) $T_1(M) = MP - PM$
- (B) $T_2(M) = M^2P - P^2M$
- (C) $T_3(M) = MP^2 + P^2M$
- (D) $T_4(M) = MP^2 - PM^2$

3.4 Groups

JAM 2026

Q.9

MCQ

[1 Mark]

Which ONE of the following statements is FALSE?

- (A) S_3 is isomorphic to a subgroup of S_4 .
- (B) $\mathbb{Z}_3$ is isomorphic to a subgroup of S_4 .
- (C) S_3 is isomorphic to a quotient group of S_4 .
- (D) $\mathbb{Z}_6$ is isomorphic to a quotient group of S_4 .

JAM 2026

Q.26

MCQ

[2 Marks]

Let $f : G \rightarrow \mathbb{Z}$ be a surjective homomorphism from abelian group G , kernel K . Which ONE is FALSE?

- (A) $\exists \phi : \mathbb{Z} \rightarrow G$ such that $f \circ \phi = \text{id}_{\mathbb{Z}}$
- (B) K is isomorphic to a quotient group of G

- (C) For all non-zero $\alpha : \mathbb{Z} \rightarrow G$ and $\beta : G \rightarrow K$, $\beta \circ \alpha \neq 0$
- (D) G has a subgroup isomorphic to $\mathbb{Z}$

JAM 2026

Q.27

MCQ

[2 Marks]

Which ONE of the following statements is FALSE?

- (A) There is a surjective group homomorphism from the additive group of rational numbers to the multiplicative group of all complex roots of unity.
- (B) The multiplicative group of complex numbers of modulus one is isomorphic to a quotient group of the additive group of real numbers.
- (C) Any group homomorphism from the multiplicative group of nonzero complex numbers into the group of all invertible 2×2 matrices with real entries has nontrivial kernel.
- (D) There exists a group homomorphism from the symmetric group on n symbols into the multiplicative group of all invertible $n \times n$ matrices with real entries, which has a trivial kernel.

JAM 2026

Q.30

MCQ

[2 Marks]

Let $G = \mathcal{P}(\{1, 2, 3, 4\})$ with $A \Delta B = (A \setminus B) \cup (B \setminus A)$. Which ONE is TRUE?

- (A) $\{1\}$ is an identity element of (G, Δ) .
- (B) (G, Δ) is an abelian group but not cyclic.
- (C) (G, Δ) is a group and has an element of order 4.
- (D) (G, Δ) is a group and has an element of order 8.

JAM 2026

Q.47

NAT

[1 Mark]

Let C_n denote a cyclic group having n elements. If there is a surjective group homomorphism from C_n to C_{30} , then the total number of such distinct surjective homomorphisms is _____.

JAM 2025

Q.9

MCQ

[1 Mark]

Let G be a finite abelian group of order 10. Let x_0 be an element of order 2 in G . Number of elements in $\{x \in G : x^3 = x_0\}$ is:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

JAM 2025

Q.13

MCQ

[2 Marks]

Let $X = \{x \in S_4 : x^3 = \text{id}\}$ and $Y = \{x \in S_4 : x^2 \neq \text{id}\}$. If $|X| = m, |Y| = n$:

- (A) m is even and n is even
- (B) m is odd and n is even
- (C) m is even and n is odd
- (D) m is odd and n is odd

JAM 2025**Q.23****MCQ****[2 Marks]**

Let m = number of injective homomorphisms from $\mathbb{Z}_3 \rightarrow \mathbb{R}/\mathbb{Z}$, and n = number of homomorphisms from $\mathbb{R}/\mathbb{Z} \rightarrow \mathbb{Z}_3$.

- (A) $m = 2, n = 1$
- (B) $m = 3, n = 3$
- (C) $m = 2, n = 3$
- (D) $m = 1, n = 1$

JAM 2025**Q.39****MSQ****[2 Marks]**

For $U(n) = \{x \in \mathbb{Z}_n : \gcd(x, n) = 1\}$:

- (A) $U(8)$ is a cyclic group
- (B) $U(5)$ is a cyclic group
- (C) $U(12)$ is a cyclic group
- (D) $U(9)$ is a cyclic group

JAM 2025**Q.52****NAT****[2 Marks]**

For $\sigma = (1\ 2\ 3) \in S_4$, $|\{\tau \in S_4 : \tau\sigma\tau^{-1} = \sigma\}| = \underline{\hspace{2cm}}$.

JAM 2025**Q.59****NAT****[2 Marks]**

Let G be an abelian group of order 35. Let m denote the number of elements of order 5 in G , and let n denote the number of elements of order 7 in G . Then $m + n = \underline{\hspace{2cm}}$.

JAM 2025**Q.60****NAT****[2 Marks]**

The number of surjective group homomorphisms from S_4 to $\mathbb{Z}_6$ is equal to $\underline{\hspace{2cm}}$.

JAM 2024**Q.5****MCQ****[1 Mark]**

Let G be a group of order 39 such that it has exactly one subgroup of order 3 and exactly one subgroup of order 13. Then, which one of the following statements is TRUE?

- (A) G is necessarily cyclic
- (B) G is abelian but need not be cyclic
- (C) G need not be abelian
- (D) G has 13 elements of order 13

JAM 2024**Q.6****MCQ****[1 Mark]**

Which one of the following is TRUE for $U(n)$ groups?

- (A) $U(5)$ is isomorphic to $U(8)$
- (B) $U(10)$ is isomorphic to $U(12)$
- (C) $U(8)$ is isomorphic to $U(10)$
- (D) $U(8)$ is isomorphic to $U(12)$

JAM 2024**Q.7****MCQ****[1 Mark]**

Which one of the following is TRUE for the symmetric group S_{13} ?

- (A) S_{13} has an element of order 42
- (B) S_{13} has no element of order 35
- (C) S_{13} has an element of order 27
- (D) S_{13} has no element of order 60

JAM 2024**Q.8****MCQ****[1 Mark]**

Let G be a finite group containing a non-identity element which is conjugate to its inverse. Then, which one of the following is TRUE?

- (A) The order of G is necessarily even
- (B) The order of G is not necessarily even
- (C) G is necessarily cyclic
- (D) G is necessarily abelian but need not be cyclic

JAM 2024**Q.11****MCQ****[2 Marks]**

Which one of the following groups has elements of order 1, 2, 3, 4, 5 but does not have an element of order greater than or equal to 6?

- (A) The alternating group A_6
- (B) The alternating group A_5
- (C) S_6
- (D) S_5

JAM 2024**Q.12****MCQ****[2 Marks]**

Consider the group $G = \{A \in M_2(\mathbb{R}) : AA^T = I_2\}$ with respect to matrix multiplication. The center $Z(G)$ has order:

- (A) 1
- (B) 2
- (C) 4
- (D) Infinite

JAM 2024**Q.33****MSQ****[2 Marks]**

Which of the following statements is/are TRUE?

- (A) The additive group of real numbers is isomorphic to the multiplicative group of positive real numbers
- (B) The multiplicative group of nonzero real numbers is isomorphic to the multiplicative group of nonzero complex numbers

- (C) The additive group of rational numbers is isomorphic to the multiplicative group of positive rational numbers
- (D) The multiplicative group of positive rational numbers is isomorphic to the multiplicative group of nonzero complex numbers

JAM 2024

Q.36

MSQ

[2 Marks]

Which of the following is/are TRUE for any group G ?

- (A) If G is non-abelian and $Z(G)$ contains more than one element, then the center of the quotient group $G/Z(G)$ contains only one element
- (B) If $|G| \geq 2$, then there exists a non-trivial homomorphism from $\mathbb{Z}$ to G
- (C) If $|G| \geq 2$ and G is non-abelian, then there exists a non-identity isomorphism from G to itself
- (D) If $|G| = p^3$, where p is a prime number, then G is necessarily abelian

JAM 2024

Q.40

MSQ

[2 Marks]

Consider $G = \{m + n\sqrt{2} : m, n \in \mathbb{Z}\}$ as a subgroup of the additive group $\mathbb{R}$. Which of the following statements is/are TRUE?

- (A) G is a cyclic subgroup of $\mathbb{R}$ under addition
- (B) $G \cap I$ is non-empty for every non-empty open interval $I \subseteq \mathbb{R}$
- (C) G is a closed subset of $\mathbb{R}$
- (D) G is isomorphic to the group $\mathbb{Z} \times \mathbb{Z}$, where $(m_1, n_1) + (m_2, n_2) = (m_1 + m_2, n_1 + n_2)$

JAM 2023

Q.1

MCQ

[1 Mark]

Let G be a finite group. Then G is necessarily a cyclic group if the order of G is:

- (A) 4
- (B) 7
- (C) 6
- (D) 10

JAM 2023

Q.23

MCQ

[2 Marks]

Consider the following statements:

- I. Every infinite group has infinitely many subgroups.
- II. There are only finitely many non-isomorphic groups of a given finite order.

Then:

- (A) both I and II are TRUE
- (B) I is TRUE but II is FALSE
- (C) I is FALSE but II is TRUE
- (D) both I and II are FALSE

JAM 2023**Q.27****MCQ****[2 Marks]**

How many group homomorphisms are there from $\mathbb{Z}_2$ to S_5 ?

- (A) 40
- (B) 41
- (C) 26
- (D) 25

JAM 2023**Q.29****MCQ****[2 Marks]**

From the additive group $(\mathbb{Q}, +)$, to which one of the following groups does there exist a non-trivial group homomorphism?

- (A) $\mathbb{R}^\times$, the multiplicative group of non-zero real numbers
- (B) $(\mathbb{Z}, +)$, the additive group of integers
- (C) $\mathbb{Z}_2$, the additive group of integers modulo 2
- (D) $\mathbb{Q}^\times$, the multiplicative group of non-zero rational numbers

JAM 2023**Q.32****MSQ****[2 Marks]**

Let $0 \in A \subseteq \mathbb{Z}$. Which of the following conditions imply that A is a subgroup of $(\mathbb{Z}, +)$?

- (A) $-2A \subseteq A, A + A = A$
- (B) $A = -A, A + 2A = A$
- (C) $A = -A, A + A = A$
- (D) $2A \subseteq A, A + A = A$

JAM 2023**Q.45****NAT****[1 Mark]**

For $\sigma \in S_8$, let $o(\sigma)$ denote the order of σ . Then $\max\{o(\sigma) : \sigma \in S_8\} = \underline{\hspace{2cm}}$.

JAM 2023**Q.46****NAT****[1 Mark]**

The number of group isomorphisms from $\mathbb{Z}_8^\times$ onto itself is $\underline{\hspace{2cm}}$.

JAM 2023**Q.54****NAT****[2 Marks]**

The number of permutations in S_4 that have exactly 2 cycles in their cycle decomposition is $\underline{\hspace{2cm}}$.

JAM 2022**Q.4****MCQ****[1 Mark]**

Let G be a group of order 2022. Let H and K be subgroups of G of order 337 and 674, respectively. If $H \cup K$ is also a subgroup of G , then which one of the following is FALSE?

- (A) H is a normal subgroup of $H \cup K$.
- (B) The order of $H \cup K$ is 1011.
- (C) The order of $H \cup K$ is 674.
- (D) K is a normal subgroup of $H \cup K$.

JAM 2022**Q.19****MCQ****[2 Marks]**

Consider the group $(\mathbb{Q}, +)$ and its subgroup $(\mathbb{Z}, +)$. For the quotient group $\mathbb{Q}/\mathbb{Z}$, which one of the following is FALSE?

- (A) $\mathbb{Q}/\mathbb{Z}$ contains a subgroup isomorphic to $(\mathbb{Z}, +)$.
- (B) There is exactly one group homomorphism from $\mathbb{Q}/\mathbb{Z}$ to $(\mathbb{Q}, +)$.
- (C) For all $n \in \mathbb{N}$, there exists $g \in \mathbb{Q}/\mathbb{Z}$ such that the order of g is n .
- (D) $\mathbb{Q}/\mathbb{Z}$ is not a cyclic group.

JAM 2022**Q.24****MCQ****[2 Marks]**

For the group $U_{37} = \{g \in \mathbb{Z}_{37} : \gcd(g, 37) = 1\}$ under multiplication modulo 37. Which ONE of the following is FALSE?

- (A) The set $\{\bar{g} \in U_{37} : \bar{g} = \bar{g}^{-1}\}$ contains exactly 2 elements.
- (B) The order of the element 10 in U_{37} is 36.
- (C) There is exactly one group homomorphism from U_{37} to $(\mathbb{Z}, +)$.
- (D) There is exactly one group homomorphism from U_{37} to $(\mathbb{Q}, +)$.

JAM 2022**Q.37****MSQ****[2 Marks]**

Let G be a finite group of order at least two. Let $\sigma : G \rightarrow G$ be a bijective group homomorphism such that $\sigma(g) = g \implies g = e$, and $\sigma \circ \sigma = \text{id}$. Which of the following is/are correct?

- (A) For each $g \in G$, there exists $h \in G$ such that $h^{-1}\sigma(h) = g$.
- (B) There exists $x \in G$ such that $x\sigma(x) \neq e$.
- (C) The map σ satisfies $\sigma(x) = x^{-1}$ for every $x \in G$.
- (D) The order of the group G is an odd number.

JAM 2022**Q.45****NAT****[1 Mark]**

The number of distinct subgroups of $\mathbb{Z}_{999}$ is _____.

JAM 2022**Q.46****NAT****[1 Mark]**

The number of elements of order 12 in the symmetric group S_7 is equal to _____.

JAM 2022**Q.58****NAT****[2 Marks]**

Let σ be the permutation in the symmetric group S_5 given by $\sigma(1) = 2, \sigma(2) = 3, \sigma(3) = 1, \sigma(4) = 5, \sigma(5) = 4$. Then the number of elements in the centralizer $N(\sigma) = \{\tau \in S_5 : \sigma \circ \tau = \tau \circ \sigma\}$ is equal to _____.

4 Master Answer Key (2022 – 2026)

JAM 2026			JAM 2025			JAM 2024			JAM 2023			JAM 2022		
Q#	Type	Key	Q#	Type	Key	Q#	Type	Key	Q#	Type	Key	Q#	Type	Key
1	MCQ	D	1	MCQ	C	1	MCQ	B	1	MCQ	B	1	MCQ	C
2	MCQ	D	2	MCQ	B	2	MCQ	A	2	MCQ	D	2	MCQ	D
3	MCQ	A	3	MCQ	A	3	MCQ	A	3	MCQ	A	3	MCQ	C
4	MCQ	D	4	MCQ	A	4	MCQ	A	4	MCQ	D	4	MCQ	B
5	MCQ	A	5	MCQ	A	5	MCQ	A	5	MCQ	A	5	MCQ	B
6	MCQ	B	6	MCQ	D	6	MCQ	D	6	MCQ	B	6	MCQ	D
7	MCQ	B	7	MCQ	B	7	MCQ	A	7	MCQ	A	7	MCQ	C
8	MCQ	B	8	MCQ	C	8	MCQ	A	8	MCQ	B	8	MCQ	A
9	MCQ	D	9	MCQ	A	9	MCQ	D	9	MCQ	C	9	MCQ	A
10	MCQ	MTA	10	MCQ	A	10	MCQ	B	10	MCQ	C	10	MCQ	D
11	MCQ	C	11	MCQ	B	11	MCQ	A	11	MCQ	B	11	MCQ	C
12	MCQ	C	12	MCQ	B	12	MCQ	B	12	MCQ	C	12	MCQ	C
13	MCQ	D	13	MCQ	B	13	MCQ	A	13	MCQ	B	13	MCQ	B
14	MCQ	D	14	MCQ	C	14	MCQ	D	14	MCQ	B	14	MCQ	D
15	MCQ	D	15	MCQ	A	15	MCQ	A	15	MCQ	D	15	MCQ	MTA
16	MCQ	C	16	MCQ	B	16	MCQ	A	16	MCQ	D	16	MCQ	A
17	MCQ	D	17	MCQ	B	17	MCQ	B	17	MCQ	B	17	MCQ	B
18	MCQ	D	18	MCQ	D	18	MCQ	A	18	MCQ	B	18	MCQ	C
19	MCQ	A	19	MCQ	D	19	MCQ	D	19	MCQ	B	19	MCQ	A
20	MCQ	A	20	MCQ	A	20	MCQ	A	20	MCQ	C	20	MCQ	C
21	MCQ	D	21	MCQ	B	21	MCQ	A	21	MCQ	B	21	MCQ	D
22	MCQ	A	22	MCQ	C	22	MCQ	D	22	MCQ	A	22	MCQ	C
23	MCQ	B	23	MCQ	A	23	MCQ	A	23	MCQ	A	23	MCQ	A
24	MCQ	C	24	MCQ	C	24	MCQ	A	24	MCQ	B	24	MCQ	B
25	MCQ	B	25	MCQ	A	25	MCQ	D	25	MCQ	A	25	MCQ	B
26	MCQ	C	26	MCQ	B	26	MCQ	B	26	MCQ	D	26	MCQ	B
27	MCQ	C	27	MCQ	C	27	MCQ	A	27	MCQ	C	27	MCQ	D
28	MCQ	D	28	MCQ	B	28	MCQ	B	28	MCQ	A	28	MCQ	B
29	MCQ	A	29	MCQ	A	29	MCQ	C	29	MCQ	A	29	MCQ	C
30	MCQ	B	30	MCQ	C	30	MCQ	A	30	MCQ	A	30	MCQ	C
31	MSQ	A,C,D	31	MSQ	A,C,D	31	MSQ	A,B	31	MSQ	A,C	31	MSQ	A,C
32	MSQ	C,D	32	MSQ	A,B,C	32	MSQ	A,B	32	MSQ	A,B,C	32	MSQ	A,B
33	MSQ	A,C	33	MSQ	A,B,D	33	MSQ	A	33	MSQ	B,D	33	MSQ	A,B,C
34	MSQ	B	34	MSQ	B,D	34	MSQ	B,D	34	MSQ	A,C,D	34	MSQ	A,D
35	MSQ	B,D	35	MSQ	A,B	35	MSQ	B,C	35	MSQ	A,B,D	35	MSQ	C,D
36	MSQ	A,C	36	MSQ	A,C,D	36	MSQ	B,C	36	MSQ	A,B,C	36	MSQ	C,D
37	MSQ	A,B,D	37	MSQ	B,D	37	MSQ	B,C	37	MSQ	A,B,C	37	MSQ	A,C,D
38	MSQ	A	38	MSQ	B,C,D	38	MSQ	B,D	38	MSQ	A,B,C	38	MSQ	A,C
39	MSQ	A,B,C	39	MSQ	B,D	39	MSQ	B,C	39	MSQ	B,D	39	MSQ	A,B
40	MSQ	B,D	40	MSQ	C,D	40	MSQ	B,D	40	MSQ	A,D	40	MSQ	A,C
41	NAT	4.0	41	NAT	2.0	41	NAT	0.25	41	NAT	0	41	NAT	2.20
42	NAT	1.0	42	NAT	4.0	42	NAT	0.20	42	NAT	3	42	NAT	14
43	NAT	1.0	43	NAT	2.0	43	NAT	1.0	43	NAT	0	43	NAT	0.33
44	NAT	1.0	44	NAT	1.41	44	NAT	0.17	44	NAT	0.50	44	NAT	0.25
45	NAT	8.0	45	NAT	3.0	45	NAT	6	45	NAT	15	45	NAT	8
46	NAT	3.0	46	NAT	2	46	NAT	3	46	NAT	6	46	NAT	420
47	NAT	8	47	NAT	4.0	47	NAT	2.0	47	NAT	0.50	47	NAT	0
48	NAT	-4.0	48	NAT	2.33	48	NAT	4.56	48	NAT	4.0	48	NAT	0.50
49	NAT	3.0	49	NAT	1	49	NAT	15	49	NAT	0.50	49	NAT	2.25
50	NAT	16.0	50	NAT	3.0	50	NAT	31	50	NAT	1.0	50	NAT	2.00
51	NAT	-1.0	51	NAT	2.0	51	NAT	6	51	NAT	21.0	51	NAT	18
52	NAT	0.25	52	NAT	3	52	NAT	0.25	52	NAT	0.75	52	NAT	4.16
53	NAT	1.25	53	NAT	9	53	NAT	2.0	53	NAT	0.50	53	NAT	3.00
54	NAT	0.45	54	NAT	5.22	54	NAT	4.80	54	NAT	11	54	NAT	2.00
55	NAT	4.0	55	NAT	1.0	55	NAT	0.33	55	NAT	0.50	55	NAT	3.00
56	NAT	-1.0	56	NAT	4.0	56	NAT	5	56	NAT	2	56	NAT	5
57	NAT	6.0	57	NAT	MTA	57	NAT	4.5	57	NAT	4	57	NAT	41
58	NAT	165	58	NAT	1	58	NAT	5	58	NAT	3	58	NAT	6
59	NAT	1.0	59	NAT	10	59	NAT	1.0	59	NAT	0	59	NAT	2.00
60	NAT	924	60	NAT	0	60	NAT	1.60	60	NAT	0.50	60	NAT	13.40